What the Stork Brought: How the Sex of a Recent Birth Shapes Mothers' Fertility Preferences

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Abstract

This paper examines how the sex of a recent birth shapes a mother's preferences over the sex composition of her children. Using data from the Demographic and Health Surveys from 1985–2022 in 82 countries covering mothers who gave birth in the past 12 months, I find that a recent daughter birth increases a mother's stated ideal number of daughters by roughly 0.11 (8.2% relative to the sample average) while decreasing her ideal number of sons by about 0.05 (3.7%); a smaller, less robust increase in the ideal number of total children (0.04, 1.2%) also emerges. Crucially, updating is not confined to the firstborn: effects at higher parities are statistically indistinguishable from first-birth effects for ideal sons and total children, and are, if anything, larger for ideal daughters. The effects on ideal daughters and sons are robust to selection on unobservables, mechanical floor effects, and a range of sensitivity analyses. The shift in preferences appears immediately after birth, remains stable over the following year, and does not extend to mothers living with another woman's newborn, a pattern more consistent with ex-post rationalization than with contact. These results imply that stated fertility preferences are endogenous to realized births, complicating their use in measuring unwanted fertility and son preference.

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1 Introduction

In 2010, an estimated 126 million women were missing from the biologically expected global sex ratio. By 2035, this number is projected to rise to 150 million (Bongaarts and Guilmoto, 2015). This phenomenon of "Missing Women", first described by Sen (1990), leaves society with an overabundance of males. Male-skewed sex ratios have negative consequences for both men and women. Surplus men face a marriage squeeze that leaves many permanently unmarried (Ebenstein and Sharygin, 2009), with elevated rates of depression, aggression, and suicidal ideation (Zhou et al., 2011). Women suffer as well: male-skewed sex ratios increase violence against women (Amaral and Bhalotra, 2017) and entrench conservative gender norms that depress female labor force participation (Grosjean and Khattar, 2019).

The rise of "Missing Women" can be attributed to son preference, where parents prefer sons to daughters (Jayachandran, 2015), but this is exacerbated by global fertility decline (Jayachandran, 2017). From 1950 to 2021, fertility rates fell from an average of 5 births per woman to 2.3 (United Nations Department of Economic and Social Affairs, 2022). This trend reflects greater individual agency in fertility decisions as a result of declining child mortality and changing values, as during Europe's earlier demographic transition (Reher et al., 2017). Son preference is also easier to act on due to increasingly available sex-selective technologies (Bhalotra and Cochrane, 2010; Kashyap and Villavicencio, 2016; Lee and Paik, 2006; Puri et al., 2011; Anukriti, Bhalotra and Tam, 2022). Because individual preferences play an ever-increasing role in the global sex ratio, and in turn in both individual and societal outcomes, it is paramount that we better understand the drivers of such preferences. In practice, these preferences cannot be observed directly, so researchers and policymakers rely heavily on stated preferences elicited in surveys (Stantcheva, 2023), and with good reason: stated fertility preferences are predictive of subsequent behavior. At the individual level, stated fertility intentions predict later births even after accounting for other characteristics (Schoen et al., 1999); at the aggregate level, desired fertility accounts for a large share of cross-country variation in realized fertility (Pritchett, 1994). Stated ideal sex composition is likewise a widely used measure of son preference across countries (Bongaarts, 2013; Jayachandran, 2017). This paper studies one driver of these preferences: the sex of a mother's most recent birth.

Having a child has been shown to change parents' preferences and behavior (Dyrdal and Lucas, 2013; Katz-Wise, Priess and Hyde, 2010; Massenkoff and Rose, 2024): the birth or death of a child causes couples to adjust fertility decisions (Shapiro and Gebreselassie, 2008), and the decision to continue to have children or not is often influenced by the sex of the child (Dahl and Moretti, 2008; Jayachandran and Pande, 2017; Lamson, 2020; Lundberg, 2005; Milazzo, 2014; Rai et al., 2014). The sex of the eldest child, in particular, has strong effects on fertility (Dahl and Moretti, 2008; Tavassoli, 2021). Dahl and Moretti (2008) estimate that 5,500 additional children were born due to the birth of a firstborn daughter each year between 1960 and 2000 in the U.S. alone. In India, eldest sons affect fertility decisions (Jayachandran and Pande, 2017), and firstborn girls lead to higher fertility and shorter spacing between births (Milazzo, 2014). That is, fertility decisions are often shaped by son preference.

Parents seek not only sons, but also sex heterogeneity among their children (Angrist and Evans, 1998; Arnold, 1997; Ben-Porath and Welch, 1976; Hank and Kohler, 2000). Parents with two children of the same sex have been observed to be more likely to have a third than parents who had one of each sex, although this trend appears to be dissipating (Ben-Porath and Welch, 1976; Pollard and Morgan, 2002; Sloane and Lee, 1983). Relatedly, Kemkes (2006) shows that a perceived surplus of daughters results in increased fertility in pursuit of more sons. A desire for a mixed-sex set of children is also observed indirectly in adoption decisions: parents are more likely to adopt a child of the missing sex than an additional child of the same sex (Gibby and Thomas, 2019). Still, women in South Korea with two daughters exhibited a stronger desire to have another child than those with two sons, indicating son preference (Yoo, Hayford and Agadjanian, 2017). Further, mixed-sex sibship preferences expand to the extended family, meaning that siblings or cousins of a couple who have a low sex mix seek to achieve balance through increased fertility (Querin, 2022). Beyond the sex composition of existing children, fertility preferences are shaped by a range of individual

characteristics, including age, education, household wealth, child mortality, and contraceptive use (Beguy and Mberu, 2015; Caldwell, 1980; Black et al., 2013; Shapiro and Gebreselassie, 2008; Ahinkorah et al., 2021).

Extensive research has been performed on how fertility preferences affect fertility outcomes, but far less is known about how births shape fertility preferences. Demographers have long recognized that women may rationalize their realized family *size*, adjusting their stated ideal upward to match the children they already have, which biases estimates of unwanted fertility downward (Bongaarts, 1990; Casterline and El-Zeini, 2007). A small literature shows that stated fertility preferences are unstable over time and respond to realized fertility and changing life circumstances (Müller et al., 2022; Sennott and Yeatman, 2012; Yeatman, Sennott and Culpepper, 2013). This work, however, focuses on desired family *size* and does not examine how the sex of a child shapes preferences over the sex *composition* of a mother's sibship. The one exception is Lamson (2020), which shows that mothers who recently gave birth to a firstborn shift their ideal numbers of sons and daughters based on the sex of the recent birth. This paper generalizes Lamson (2020)'s result beyond the firstborn, asking the following: Does the sex of a recent birth affect a mother's ideal number of daughters/sons/total children?

Existing theory points to two primary channels through which mothers' sibship composition preferences may be altered by the sex of a recent birth: ex-post rationalization and contact. The theory of ex-post rationalization is grounded in the theory of cognitive dissonance. First proposed by Festinger (1957) and introduced to economics by Akerlof and Dickens (1982), the theory of cognitive dissonance posits that when our actions or experiences are not consistent with our beliefs, we update our beliefs to be consistent with our actions to minimize psychological discomfort. Beliefs and desires produce actions, but actions produce beliefs and desires as well. Believing that we make correct decisions is comforting, and these beliefs often serve psychological needs such as self-esteem and anxiety reduction (Oeberst and Imhoff, 2023; Bénabou and Tirole, 2016). Recent work formalizes how people rationalize past choices, even mistaken ones, such that sunk costs shape subsequent decisions (Eyster, Li and Ridout, 2021), and argues that such rationalization can be adaptive, improving subsequent reasoning (Cushman, 2020). When parents have a daughter, ex-post rationalization offers a way to reconcile an outcome they did not choose with preferences they may have held for a son. If ex-post rationalization is the primary mechanism which is shifting mothers' preferences, then we would expect an immediate and stable shift of preferences in the direction of the sex of the recent birth that does not grow or fade with time.[1]

Contact theory—which posits that sustained exposure to members of a group reduces prejudice toward that group (Pettigrew and Tropp, 2006; Lowe, 2025)—offers an alternative account of why mothers' preferences might shift following the birth of a daughter. Within the household, a child of a given sex provides repeated and intimate contact with that sex. Evidence on how daughters affect their parents is consistent with this mechanism: conditional on family size, having more daughters shifts parents' political views and votes in a more liberal direction, particularly on women's issues (Washington, 2008; Oswald and Powdthavee, 2010). Applied to fertility preferences, mothers who recently gave birth to a daughter may, through day-to-day contact with her, come to place greater value on daughters, such that their ideal number of daughters rises relative to mothers who had a son. Unlike ex-post rationalization, however, contact operates through accumulated exposure rather than through a one-time revision of beliefs. If contact is the primary mechanism, we would expect the shift in preferences to be modest immediately after birth and to grow with time since birth as exposure to the child accumulates.

Based on extant theoretical underpinnings, I hypothesize that the desired total number of daughters will be higher in mothers with recently born daughters than those with recently born sons. Second, I hypothesize that the desired total number of sons will be lower in mothers with recently born daughters than those with recently born sons. Lastly, I hypothesize that the desired total number of children will be higher in mothers with recently born daughters than those

1 Müller et al. (2022), for one, highlights ex-post rationalization as a possible rationale for their findings.

with recently born sons.

The results of this paper are broadly consistent with the three aforementioned hypotheses: the effects on ideal daughters and ideal sons are strong and robust, while the effect on ideal total children is positive but fragile. Relative to mothers with a recent son birth, a recent daughter birth increases a mother's ideal number of daughters by roughly 0.11 (8.2% relative to the sample average), decreases a mother's ideal number of sons by about 0.05 (3.7%), and increases the ideal number of children by roughly 0.04 (1.2%). These observed changes in preferences, especially on a mother's stated ideal number of daughters, are stable across a battery of individual characteristics and cultural settings. While a few factors moderate the effect on ideal sons, effects on ideal daughters show no robust heterogeneity, and effects remain directionally homogeneous across all interactions tested. Finally, investigations into the mechanisms driving these observed effects show that they are more consistent with ex-post rationalization than with contact.

This paper makes three primary contributions. First, it extends Lamson (2020)'s firstborn result to births at higher parities. Because the sex of later-born children is typically treated as endogenous, existing work has relied on firstborn sex for identification and has therefore not investigated whether preferences on sibship sex composition continue to move after the first birth. Exploiting the quasi-random sex of the most recent birth within pre-existing sibship cells across 82 countries, I show that they do: effects at higher parities are statistically indistinguishable from first-birth effects for ideal sons and ideal total children and, if anything, larger for ideal daughters. Sex preferences are therefore not set at the first birth but revised at each successive birth. Second, whereas Müller et al. (2022) shows that desired sibship *size* rises with realized fertility, I show that desired sibship *composition* also moves toward the composition a mother has realized at each birth. Third, these results matter for how son preference is measured and how it evolves. If stated ideals partly reflect the children a mother already has, survey-based measures will understate son preference precisely among mothers with daughters, the group most at risk of sex selection at the next birth (Jha et al., 2011). Conversely, if the shift reflects a genuine change in preferences, realized daughters attenuate son preference over time, implying that policies which curb sex selection may yield a further dividend by reshaping the preferences that drive it.

The remainder of the paper is structured as follows: First, I lay out my data sources and methodology in detail, outlining limitations and potential sources of endogeneity. Then, I present my results complete with a multitude of sensitivity analyses. Following the main results and robustness checks, I investigate possible mechanisms and discuss the findings. I conclude with limitations and closing remarks.

2 Data and Setting

Data for this paper are sourced from the Demographic and Health Surveys (DHS) from annual waves from 1985 to 2022, containing responses and demographic characteristics on 310,007 mothers who have given birth in the past 12 months and have three or fewer children. Because some respondents give non-numeric answers to the ideal-family questions or are missing control variables, the main regressions use roughly 220,000 observations for ideal daughters and sons and 260,000 for ideal total children (Table 3). I combine this dataset with data from the UN Department of Economic and Social Affairs on country-level statistics on fertility and World Bank Indicators on country-level macroeconomic and cultural indicators. My three outcome variables are a mother's expressed ideal number of daughters, sons, and total children. These variables are constructed from two DHS questions in the women's survey.[2] The treatment

2 The DHS elicits ideal family size and sex composition through a two-part sequence in the Woman's Questionnaire. Respondents with no living children are first asked how many children they would choose to have in their lifetime if the choice were entirely theirs; those who already have children are asked the same question framed retrospectively, as if returning to a time before they had any children. Respondents who give a numeric answer are then asked how many of that total they would want to be boys, how many girls, and how many for whom the sex would not matter. Interviewers record these as three separate counts, which together should sum to the ideal number reported in the first question; respondents who instead give a non-numeric answer (e.g., "it's up to God") to either question are coded as missing rather than assigned a numeric value. To avoid outliers and outlandish and unrealistic claims, ideal total children is top-coded at 12, and ideal daughters/sons are each top-coded at 6.

variable is a dummy variable representing whether or not the mother birthed a daughter in the last 12 months. The dataset contains a number of pertinent demographic indicators, variables outlining birth order and family composition, and variables detailing where and when the responses of each individual were collected.

Descriptive statistics are shown in Table 1. The average SRB of a given country in a given year is 1.05 (104.9 male births per 100 female births). Aggregated over the entire sample at large, the SRB is 104.796 males to 100 females. Both of these measures lie within the biologically expected SRB, which hovers between 103 and 107 males per 100 females (Orzack et al., 2015). Afghanistan has the highest in-sample SRB of 119.3 males per 100 females, and Gabon has the lowest with 86.1 males per 100 females. While these skewed SRBs may seem problematic, the sample sizes in both countries are quite small.

Table 1: Summary Statistics

	mean	sd	min	max	count
Ideal Daughters	1.564	1.168	0	6	245,143
Ideal Sons	1.652	1.255	0	6	245,142
Ideal Total Children	3.741	2.064	0	12	289,099
Recent Daughter Birth	0.490	0.500	0	1	310,007
Age	24.082	4.952	13	50	310,007
Age at First Birth	20.270	3.995	12	48	309,491
Education (Years)	6.035	4.816	0	24	307,791
Rural Household	0.618	0.486	0	1	310,007
Poor Household	0.409	0.492	0	1	310,007
Unmarried	0.077	0.266	0	1	309,999
Widowed	0.153	0.360	0	1	309,999
Divorced	0.011	0.106	0	1	309,999
Age at First Marriage	18.890	3.986	12	44	284,668
Total child mortality	0.119	0.357	0	3	310,007
Sibship: First Child Only	0.403	0.490	0	1	310,007
Sibship: Son	0.174	0.379	0	1	310,007
Sibship: Daughter	0.166	0.372	0	1	310,007
Sibship: Son-Son	0.067	0.250	0	1	310,007
Sibship: Son-Daughter	0.064	0.244	0	1	310,007
Sibship: Daughter-Son	0.063	0.243	0	1	310,007
Sibship: Daughter-Daughter	0.064	0.244	0	1	310,007
High SRB Country	0.193	0.395	0	1	310,007
Total Daughters	0.909	0.789	0	3	310,007
Total Sons	0.946	0.792	0	3	310,007
Total Children	1.855	0.799	1	3	310,007
Observations	310007				

Notes: Summary statistics for the sibship design analysis sample: women with a recent birth (within the past 12 months) and $\leq$3 total children, drawn from 82 DHS countries spanning Sub-Saharan Africa, Asia, Latin America, Central Asia, and MENA. Sibship: First Child Only equals 1 for mothers whose recent birth was their first, so the sample includes both first-time mothers and mothers with pre-existing sibships. Figures reported here are unweighted; population-weighted statistics discussed in the text (which reflect global rather than sample-share patterns) will differ accordingly. The recent-daughter-birth and recent-son-birth indicators equal 1 if the most recent birth was, respectively, female or male. Education, rural, and wealth (poor) are from the DHS women's questionnaire. High SRB Country equals 1 if the country's sex ratio at birth in the analysis sample exceeds 1.07 (107 males per 100 females), the upper bound of the natural range. Also reported: marital status (unmarried/widowed/divorced, married omitted), age at first marriage, child mortality count.

On average, mothers in the sample indicate that they would ideally have 1.30 daughters, 1.49 sons, and 3.13 total children (population-weighted; unweighted figures are reported in Table 1). The discrepancy between the ideal number of daughters and sons indicates persistent son preference across the sample. 48.6% of the recent births were female (the remainder male), consistent with the natural SRB. The average age of a mother in the sample is 23.89, with the

average age of first birth occurring at 20.20. There is a wide age range, with the youngest mother of a recent birth being 13 and the oldest being 50. Mothers possess an average of 5.26 years of education and are mostly living in rural areas. Many are poor, with 18.4% falling in the bottom two DHS wealth quintiles.

Mothers have a variety of pre-existing sibships. 40.6% of mothers had no previous children prior to their recent birth, 34.6% had one child previously, and 24.8% gave birth twice previously. The possible birth orders are relatively balanced, although there are subtle deviations. Further breakdown of pre-existing sibships can be found in Table 1.

3 Methodology

3.1 Estimation

To test these hypotheses, I exploit the quasi-random sex of a mother's most recent birth to estimate its effect on her stated preferences. My primary specification to analyze treatment effects uses ordinary least squares regression (OLS). Following Lamson (2020), I adapt her OLS specification to a more general case. I include fixed effects for both interview year and interview month interacted with the administrative region, along with heteroskedasticity-consistent standard errors. The OLS model specification is outlined as follows:

$$P_{idy} = \alpha + \tilde{\beta} D_{idy} + X'_{idy}\theta + \phi_{dm} + \gamma_y + u_{idy} \tag{1}$$

The outcome variable P_{idy} is the ideal number of daughters, sons, or total children desired for mother i in region d in year y. The treatment variable D_{idy} is a dummy equal to one if mother i had a daughter birth in the past 12 months, and zero if they had a son (given that I restrict my sample to only mothers with recent births). X_{idy} is a vector of controls including the mother's age, age at first birth, years of education, dummies for rural and poor households, marital status (unmarried/widowed/divorced, married omitted), age at first marriage, child mortality count, and pre-existing sibship indicators prior to the recent birth, with θ the corresponding coefficient vector. Fixed effects are indicated by ϕ_{dm} and γ_y, representing fixed effects for interview month interacted with administrative region, as well as fixed effects for interview year. Observations are weighted by each respondent's DHS sample weight, normalized by survey size, scaled by her country's population in 2000, and divided by the number of surveys from her country, so that each country contributes in proportion to its population regardless of how many surveys it fielded; 2000 lies near the middle of the sample period.[3]

3.2 Identification

The identification strategy of this paper rests on the sex of a mother's most recent birth being quasi-randomly assigned, conditional on her pre-existing sibship. This source of exogeneity faces three distinct threats: (1) sex selection at birth, (2) survival selection between birth and interview, and (3) sample or response selection into the analysis sample itself. I address each in turn.

3.2.1 Threat 1: Sex Selection at Birth

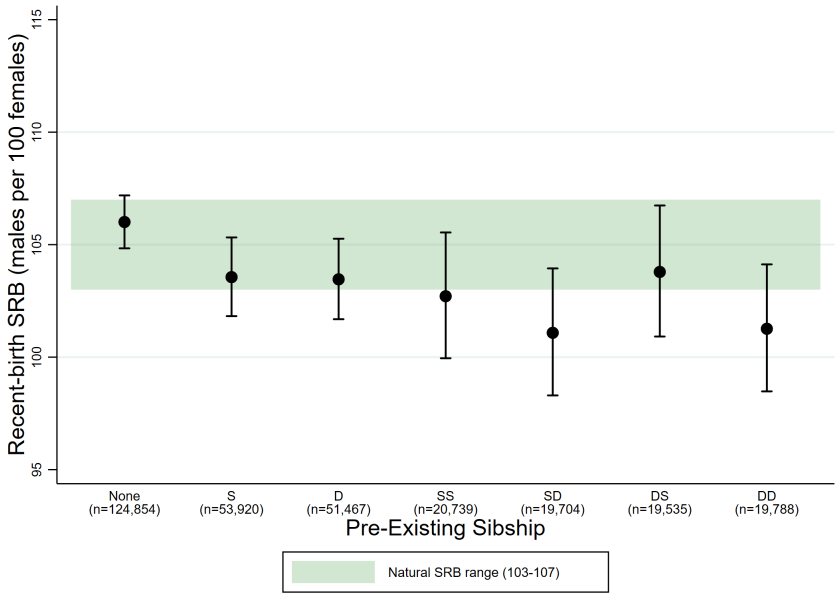
While many well-identified studies have used the sex of the firstborn child as their source of randomization (Ananat and Michaels, 2008; Bedard and Deschenes, 2005; Lamson, 2020; Lee, 2008), the sex of a child at birth is endogenous with previous births beyond the firstborn due to its effects on parental preferences (Bhalotra and Cochrane, 2010; Lundberg, 2005). In some regions of India and China, the problem of "Missing Women" is severe, peaking at around 115 males per 100 females (Anderson and Ray, 2010; Anttila-Hughes, Krause and Stopnitzky, 2016; Coale, 1991; Das Gupta

3 As robustness checks to my OLS estimates, I also estimate treatment effects using semi-parametric methods: propensity score matching and inverse probability weighting with regression adjustment (IPWRA) estimators in Appendix sections A.13 and A.14 respectively.

et al., 2003; Sen, 1990). Michael et al. (2013) examines an even more extreme case, finding evidence for sex-selective abortion at a national scale in Armenia in the second born, where SRBs of the second born if the firstborn is a daughter skew to 156 males per 100 females.

If there is indeed persistent skewing towards sons across the sample at large, we would see departures from the natural SRB boundaries, especially in higher birth orders. As seen in Figure 1, this is not the case. Using SRB estimates from the survey itself combined with the Wald test to generate confidence intervals, none of the previous sibship samples are systematically distinguishable from the natural SRB rate of 103–107 males per 100 females. Consistent with this, Table A.1 shows that the sex of the recent birth is statistically orthogonal to prior family composition.

Figure 1: Recent-Birth SRB by Pre-Existing Sibship

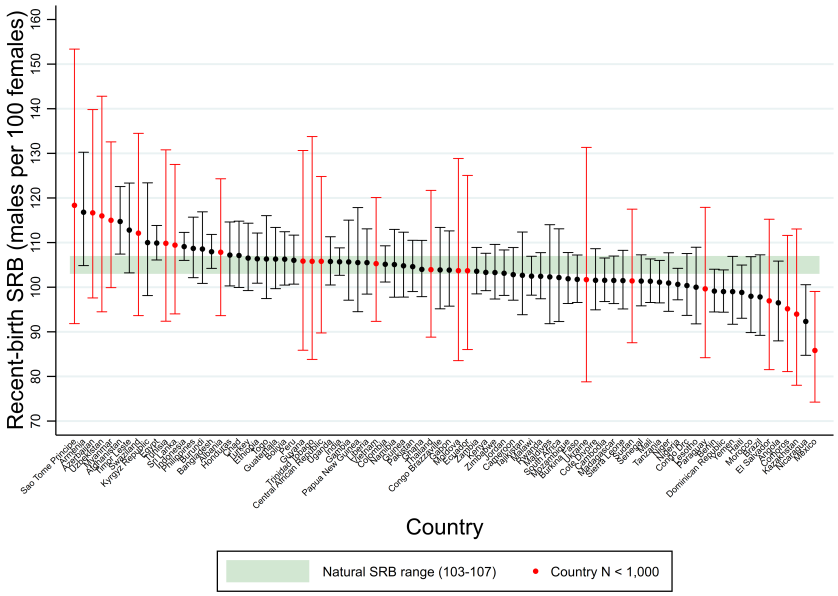


Notes: Point = observed SRB (males per 100 females) among recent births by pre-existing sibship cell. Caps = 95% CIs, calculated using the Wald test. Green band = natural SRB range (103–107). Deviations outside this range would indicate sex-selective behavior; all cells are consistent with a natural SRB.

Figure 2 extends this test across all 82 countries individually rather than by sibship cell. Only 3 of the 82 countries have 95% confidence intervals falling entirely outside the natural 103–107 range. 22 of the 82 countries (27%) have fewer than 1,000 usable observations (shown in red in the figure), so their point estimates are noisier and less reliable than the rest of the sample.

In addition to the apparent absence of systematic SRB skewing in higher birth orders, maternal covariates are also largely balanced when splitting the sample by mothers of recent daughters and of recent sons in Table 2. Five covariates do remain unbalanced at the 5% level, however: a son is more likely to be a mother's first birth, widowed mothers are more likely to have birthed a daughter, mothers who experienced higher levels of child mortality are more likely to have birthed a son, and two of the pre-existing sibship-composition dummies (Son-Daughter and Daughter-Daughter) are also imbalanced. That the sibship-composition imbalances concentrate in the pre-existing family structure is precisely why the primary specification controls for sibship cell: within a given cell, the sex of the recent birth is orthogonal to prior family composition by construction (Table A.1), so cell-level imbalance in the pooled sample does not threaten identification of the within-cell treatment effect. Figure A.1 further shows the propensity score distributions across recent-daughter and recent-son mothers are nearly identical, indicating plenty of common support and an appropriate environment to employ matching estimators.

Figure 2: Recent-Birth SRB by Country



Notes: Observed SRB (males per 100 females) among recent births, computed separately by country and sorted from highest to lowest. Point = country-level SRB; capped lines = 95% Wald CIs; green band = natural SRB range (103–107). Countries with fewer than 1,000 usable observations are shown in red (point and confidence interval) since their SRB estimates are noisy; all other countries are shown in black. Constructed analogously to Figure 1, which splits by pre-existing sibship cell instead of country.

Table 2: Covariate Balance

	Son/Control	Daughter/Treated	Diff.	t
Age	24.080	24.083	-0.003	-0.17
Age at First Birth	20.279	20.262	0.017	1.17
Education (Years)	6.045	6.024	0.020	1.18
Rural Household	0.618	0.617	0.001	0.58
Poor Household	0.411	0.408	0.002	1.28
Interview Month	6.861	6.846	0.014	1.16
Interview Year	2006.119	2006.101	0.019	0.54
Sibship: First Child Only	0.406	0.399	0.007***	4.00
Sibship: Son	0.173	0.174	-0.001	-0.75
Sibship: Daughter	0.165	0.167	-0.001	-0.85
Sibship: Son-Son	0.066	0.067	-0.001	-1.05
Sibship: Son-Daughter	0.063	0.065	-0.002**	-2.18
Sibship: Daughter-Son	0.063	0.063	-0.000	-0.26
Sibship: Daughter-Daughter	0.063	0.065	-0.002**	-2.06
Unmarried	0.077	0.076	0.001	1.01
Widowed	0.152	0.155	-0.003**	-2.26
Divorced	0.011	0.011	-0.000	-0.72
Age at First Marriage	18.895	18.886	0.009	0.60
Total child mortality	0.122	0.117	0.005***	3.91
Observations	310,007			

Notes: t-tests comparing women with a recent son birth (control) vs. recent daughter birth (treated), on the full covariate set used in the OLS specifications plus interview month/year and sibship dummies (i.e., the PSM matching covariates augmented with marital status, age at first marriage, child mortality). Sample: women with a recent birth ($\leq$12 months) and $\leq$3 total children. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

SRBs fall within the natural range in nearly every sibship cell and country, and covariates are balanced apart from the five exceptions above, so the sex of the recent birth is close to as-good-as-random both across and within sibship cells.

3.2.2 Threat 2: Survival Selection

Mothers in this paper are not sampled directly after the birth of a child, so it is also imperative to consider the implications of other factors affecting sex ratios amongst the recently birthed. Namely, the key, tragic phenomena in this regard are early childhood abandonment, sex-selective childcare, and female infanticide (Barcellos, Carvalho and Lleras-Muney, 2010; Bongaarts and Guilmoto, 2015; Hesketh and Xing, 2006; Miller, 1987). While these do not affect sex ratio at birth, they may affect the sex ratio at time of survey if mothers are not wholly truthful about prior deaths of children, or if a daughter born 12 months ago did not survive to be reported as a "recent birth."

Table 2 shows that mothers who experienced higher levels of child mortality are significantly more likely to have birthed a son, which is consistent with sex-selective childcare or female infanticide operating somewhere in this sample; this possibility is investigated further in Section 4.3.2.

Treatment, sibship-cell, and total-children variables throughout this paper are constructed from each mother's full retrospective birth history—including children who have since died—rather than from her currently-living children. Sex at birth is therefore assigned intent-to-treat regardless of subsequent survival, which directly neutralizes the concern that the sex ratio at time of survey may diverge from the sex ratio at birth.

3.2.3 Threat 3: Sample and Response Selection

Conditioning on "a birth in the last 12 months" conditions on a behavior—continued childbearing—that is itself affected by prior child sex: son-biased stopping rules mean that mothers who reach parity 3 with two daughters are a selected group. Conditioning on sibship cell handles composition *across* cells, but cell membership itself is potentially endogenous.

If son-biased stopping rules shape who continues childbearing, the pre-existing sibship-cell shares observed in this sample should deviate from what pure chance would predict. Using the sample's observed recent-birth male share (51.3%) as the per-birth probability, the four two-prior-child cells (Son-Son, Son-Daughter, Daughter-Son, Daughter-Daughter) are expected to occur with probability p^2, $p(1-p)$, $(1-p)p$, and $(1-p)^2$ respectively. A χ^2 goodness-of-fit test rejects this binomial benchmark decisively ($\chi^2(3) = 56.02$, $p < 0.001$): Daughter-Daughter sibships are overrepresented (6.4% observed vs. 6.1% expected) while Son-Son and Daughter-Son sibships are each modestly underrepresented (6.7% vs. 6.8%; 6.3% vs. 6.4%), consistent with mothers who already have two daughters being more likely than chance to continue childbearing in pursuit of a son. By contrast, the one-prior-child cells (Son vs. Daughter) match the binomial benchmark almost exactly, and an analogous test cannot reject it ($p = 0.28$). This confirms the endogenous-selection concern is empirically present in some previous sibship samples.

This is precisely why the primary specification conditions on pre-existing sibship cell: the identifying variation is the quasi-random sex of the most recent birth within a cell, not the composition of cells themselves, so endogenous selection into a cell does not contaminate the treatment effect. Separately, the estimand throughout this paper is a treatment effect local to mothers who have recently given birth, not a population-average effect across all mothers of reproductive age. This restriction is a feature rather than a limitation of the design: preference updating in response to a birth can only be measured among mothers who have just experienced one, so the "recently-birthed" population is precisely the population of interest for the question this paper asks.

3.3 Oster Bounds

Oster (2019) bounds provide a means of assessing the sensitivity of OLS estimates to omitted variable bias, quantifying

the degree of unobservable selection that would be required to reduce a coefficient to zero. Consider a theoretical case where we are able to include all unobservables, described as O_{idy}, in the primary OLS specification:

$$P_{idy} = \alpha + \beta D_{idy} + X'_{idy}\theta + O_{idy} + \phi_d + \gamma_y + u_{idy} \tag{2}$$

Define δ as the degree of selection on unobservables relative to selection on observables. A parsimonious regression including only the treatment is then estimated on the same sample and with the same population weights as Equation 1:

$$P_{idy} = \alpha + \breve{\beta} D_{idy} + u_{idy} \tag{3}$$

Define $\breve{\beta}$ as the coefficient on D_{idy} in Equation 3, and $\breve{R}^2$ as the R^2 in Equation 3. Similarly, define $\tilde{\beta}$ as the coefficient on D_{idy} in the original specification Equation 1, and $\tilde{R}^2$ as the R^2 in Equation 1.

Finally, define $R_{\max}$ as the theoretical R^2 in Equation 2. This cannot be estimated, so an assumption about its value is required. Following Oster (2019), I set $R_{\max} = 1.3\tilde{R}^2$, and I report results using $R_{\max} = \tilde{R}^2 + 0.1$ as a sensitivity check.

Oster shows that, for the true β of theoretical Equation 2 to equal 0, $\hat{\delta}$ can be estimated—the value for δ that would mean that D_{idy} actually has a null effect on P_{idy}—using the following equation:

$$\hat{\delta} = \frac{\tilde{\beta}(\tilde{R}^2 - \breve{R}^2)}{(\breve{\beta} - \tilde{\beta})(R_{\max} - \tilde{R}^2)} \tag{4}$$

A value of $\hat{\delta}$ greater than one indicates that unobservables would need to be more important than all included observables to drive the true effect to zero, the conventional robustness threshold (Oster, 2019). A negative $\hat{\delta}$ arises when adding controls moves the estimate away from zero; selection on unobservables proportional to selection on observables would then push the estimate further from zero, so the result is likewise robust. As a complementary measure, I also report the bias-adjusted coefficient β^*, obtained by setting $\delta = 1$:

$$\beta^* = \tilde{\beta} - (\breve{\beta} - \tilde{\beta})\frac{R_{\max} - \tilde{R}^2}{\tilde{R}^2 - \breve{R}^2} \tag{5}$$

If the interval between $\tilde{\beta}$ and β^* excludes zero, the estimate is robust to selection on unobservables as strong as selection on observables.

4 Results

To reiterate, I hypothesize that a recent daughter birth will have an effect on a mother's stated ideal number of daughters, sons, and total children. Results indicate that a recent daughter birth increases a mother's ideal number of daughters by about 0.11, decreases her ideal number of sons by about 0.05, and increases her ideal number of children by about 0.04. In particular, the effect of a recent daughter birth on a mother's ideal number of daughters is exceedingly robust to high levels of potential endogeneity, is apparent in all sub-sample analyses, and has almost no heterogeneous treatment effects. The negative effect of the treatment of a recent daughter birth on a mother's ideal sons is persistent throughout robustness checks and sensitivity analyses as well, but to a lesser magnitude and significance. The positive effect of a recent daughter birth on a mother's ideal number of total children is less robust, with noisy estimates within individual sibship cells, directionally heterogeneous treatment effects across some sub-samples, and an insignificant matching estimate.

4.1 Primary OLS Specification Results

Table 3 presents my primary OLS specification results. Odd numbered specifications ((1), (3), and (5)) are parsimonious specifications, containing no controls or fixed effects and estimated on the same sample and with the same population weights as the full specifications. These three simple specifications estimate that a recent daughter birth increases a mother's ideal number of daughters by 0.1224 at a 1% significance level, decreases a mother's ideal number of sons by -0.0388 at a 1% significance level, and increases a mother's ideal number of total children by 0.0584 at a 1% significance level.

Table 3: Primary OLS Results

	Ideal Daughters		Ideal Sons		Ideal Total Children	
	(1)	(2)	(3)	(4)	(5)	(6)
	Parsim.	Full	Parsim.	Full	Parsim.	Full
Recent Daughter Birth	0.1224***	0.1070***	-0.0388***	-0.0549***	0.0584***	0.0381***
	(0.0087)	(0.0069)	(0.0101)	(0.0081)	(0.0127)	(0.0094)
Constant	1.2298***	1.3241***	1.5087***	1.6551***	3.1003***	3.2482***
	(0.0059)	(0.0235)	(0.0070)	(0.0269)	(0.0088)	(0.0339)
Oster $\hat{\delta}$ ($R_{\max} = 1.3\tilde{R}^2$)		22.874		-11.364		6.245
Oster β^* ($\delta = 1$)		0.1023		-0.0597		0.0320
Observations	219,991	219,991	219,990	219,990	260,438	260,438
R-squared	0.0035	0.3540	0.0003	0.3607	0.0003	0.5277

Notes: Dependent variables: stated ideal number of daughters, sons, and total children. Treatment: sex of most recent birth in the past 12 months. "Full" specification controls for sibship composition dummies, age, age at first birth, education (years), rural, poor, marital status (unmarried/widowed/divorced, married omitted), age at first marriage, child mortality count; absorbs region×interview-month and interview-year fixed effects; population-weighted. Parsimonious specifications are estimated on the full-specification sample with the same weights. Oster $\hat{\delta}$ uses $R_{\max} = 1.3\tilde{R}^2$; β^* is the bias-adjusted coefficient at $\delta = 1$ (Equation 5). Robust SEs. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Specifications (2), (4), and (6) in Table 3 have the complete set of controls and fixed effects as described in Section 3.1. Adding controls moves the estimates somewhat—toward zero for ideal daughters and ideal total children, and away from zero for ideal sons—but all remain significant at the 1% level and maintain the directionality from the parsimonious specifications.

Oster bounds support a robust reading of all three estimates. For ideal daughters, adding controls moves the estimate modestly toward zero (from 0.1224 to 0.1070), yielding $\hat{\delta} = 22.874$: unobservables would need to be roughly 22.874 times as important as the included observables to eliminate the effect. For ideal sons, adding controls moves the estimate away from zero (from -0.0388 to -0.0549), so $\hat{\delta}$ is negative (-11.364), and selection on unobservables proportional to selection on observables would only strengthen the effect. For ideal total children, controls move the estimate toward zero (from 0.0584 to 0.0381), and $\hat{\delta} = 6.245$ remains well above one. In all three cases, the bias-adjusted coefficient β^* (0.1023, -0.0597, and 0.0320, respectively) shares the sign of the full-specification estimate, so the interval between $\tilde{\beta}$ and β^* excludes zero. These conclusions are unchanged when setting $R_{\max} = \tilde{R}^2 + 0.1$ ($\hat{\delta}$ of 24.289, -12.297, and 9.887) or when the baseline regression includes the region×interview-month and interview-year fixed effects ($\hat{\delta}$ of 15.740, -3.600, and 2.278). I read the total-children result cautiously, however: it is markedly less robust than the other two by every other diagnostic in this paper, so I do not treat the Oster bound alone as sufficient grounds to trust this particular estimate.[4] Given that treatment here is plausibly close to randomly assigned within sibship cell (Section 3.2), these bounds are best read as a belt-and-suspenders check layered on top of the design-based identification argument, rather than as the primary basis for a causal claim.

4 Only 2 of seven sibship-cell specifications are significant (Section 4.2), and the PSM average treatment effect on the treated is small, insignificant, and of the opposite sign (Table A.18)

4.2 Updating Beyond the Firstborn

Lamson (2020) finds that a recent firstborn daughter affects a mother's preferences. In particular, she finds that a firstborn daughter causes mothers to prefer a higher ideal number of daughters, a lower ideal number of sons, and a higher ideal number of total children. These findings are in line with my larger sample which measures the effects of a recent daughter birth for mothers who have had two or fewer children previously.

The question remains whether the first birth is driving mothers' preferences, or whether mothers are continuously updating their preferences with each recent birth. To answer this question, I employ my primary OLS specification with only the "no pre-existing sibship" dummy variable included out of the seven sibship dummies, and also interacting this dummy with the treatment variable (recent daughter birth). This means that the coefficient on recent daughter birth will be the estimate of treatment effect on mothers with already extant sibships prior to the recent birth, and the interaction term will represent the difference of treatment effect between mothers whose recent birth is their first birth and mothers who already have a pre-existing sibship.

Table A.2 shows the results of these specifications. For mothers with already pre-existing sibships, the treatment effect of recent daughter birth on her ideal number of daughters is 0.122, and this is significant beyond the 1% level. The coefficient on the interaction term is -0.030 and significant at the 5% level, suggesting that the treatment effect is significantly weaker among mothers with first births than mothers with pre-existing sibships.

A recent daughter birth does not have statistically significant heterogeneous treatment effects on the ideal number of sons or total children for a mother. The treatment effect of a recent daughter birth for mothers with pre-existing sibships on her ideal number of sons is -0.060, and this is significant at the 1% level, and there is no significant difference in this treatment effect and the treatment effect on mothers whose recent birth is their first birth. Similarly, the treatment effect of a recent daughter birth for mothers with pre-existing sibships on her ideal number of total children is 0.041, and this is significant at the 1% level, and there is no significant difference in this treatment effect and the treatment effect on mothers whose recent birth is their first birth. Based on these estimations, the firstborn is not the sole driver of a mother's preferences.

To further examine the differences in treatment effects of a recent daughter birth in different subsamples, I divide my data into seven subsamples.[5] The specification used on these samples follows Equation 1 but controls only for age, age at first birth, education, rural, and poor households, with the same fixed effects and weights.

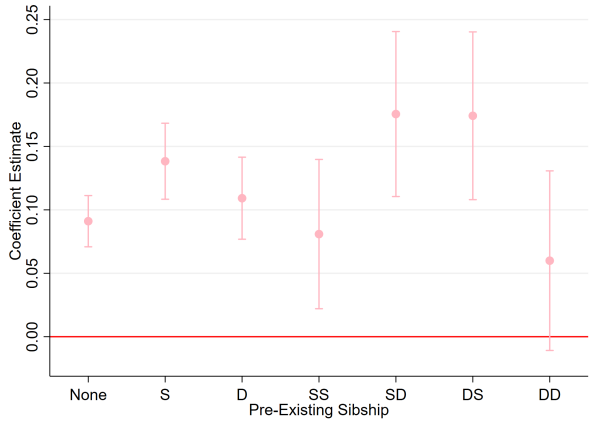
Table A.3 shows estimates of the seven specifications with a mother's ideal number of daughters as the dependent variable, and Panel (a) of Figure 3 shows these results graphically. All coefficients on recent daughter birth are directionally positive across the seven sibship-cell specifications; 6 of 7 are significant beyond or at the 5% level, and 6 of 7 are significant beyond the 1% level. There are no clear trends of treatment effects across these subsamples; they generally hover around 0.1. Standard errors increase as the sibship sizes increase, and this can be attributed to the decrease in sample size. It is also possible that sex-selective abortion or early childhood abandonment is increasing as the number of children increases, making the results less robust.

Table A.4 displays results for when a mother's ideal number of sons is the dependent variable, with Panel (b) of Figure 3 highlighting these results. The results are far less robust than when the ideal number of daughters is the dependent variable, with only 4 of 7 specifications resulting in treatment effects significant beyond the 10% level. All treatment effects estimations are negative, even though they are not all significant. Standard errors again grow as sample size decreases, and presumably the relatively lower magnitude of the treatment effects with the ideal number of sons as the dependent variable makes statistical significance harder to achieve (as compared with the ideal number

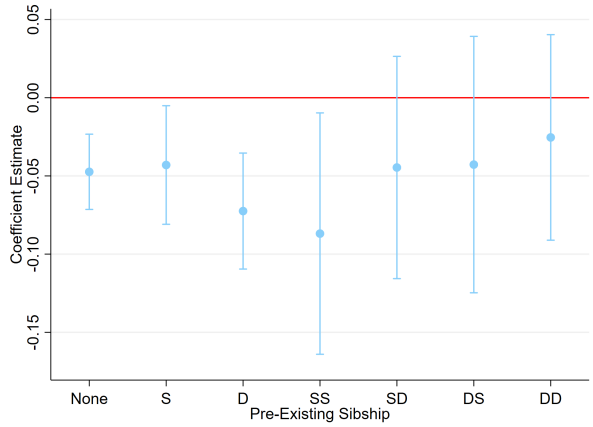
5 Each subsample is defined by the mother's births before the recent one, in birth order (S = son, D = daughter), and corresponds to specifications (1)–(7) in Tables A.3–A.5: (1) no prior children; (2) S; (3) D; (4) SS; (5) SD; (6) DS; (7) DD.

Figure 3: By-Sibship Coefficients

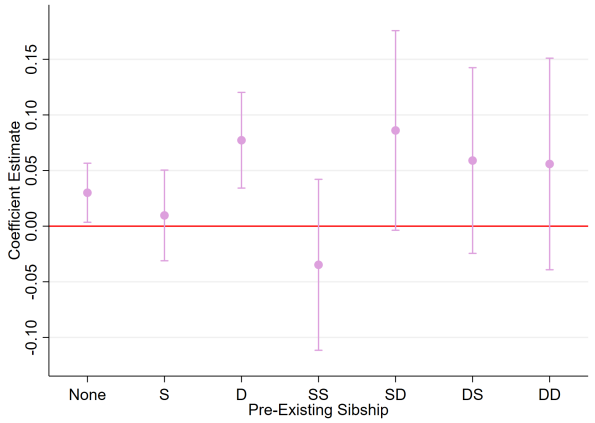
(a) Ideal Daughters



(b) Ideal Sons



(c) Ideal Total Children



Notes: Point estimates (circles) and 95% CIs (capped lines) for the effect of a recent daughter birth on stated ideal daughters, sons, and total children (Panels (a)–(c)), estimated separately within each pre-existing sibship cell (see Tables A.3–A.5). Each point on the *x*-axis labels a sibship cell: single child (1st only), then sequences of prior births in birth-rank order (S = prior son, D = prior daughter). The identifying variation within each cell is the quasi-random sex of the most recent birth. Specification: age, age at first birth, education, rural, poor; region×interview-month and interview-year FE; population-weighted; robust SEs.

of daughters specifications).

Table A.5 gives insight into results when the ideal number of total children is the dependent variable, and Panel (c) of Figure 3 graphs the treatment effects. In this group of specifications, only 2 of 7 specifications possess significant (5%) treatment effects. In this case, specification (3) is an outlier, where we see that having a recent daughter birth after the previous birth of a daughter causes mothers to want an ideal total number of children 0.077 higher than mothers with a firstborn daughter and a recent son. The treatment effect among this group is considerably higher and more significant than the other specifications in Table A.5. Further investigation into the treatment effect in this scenario is an avenue for future research, as this subgroup alone seems to partially drive the treatment effect estimations at the level of the whole sample. Specification (1) also has a positive estimated treatment effect of 0.030, significant at the 5% level. This first-birth estimate is smaller than the pooled effect among mothers with earlier children (0.041; Table A.2), so the firstborn does not drive the total-children result, although estimates within individual cells are noisy.

In summary, the firstborn is not the sole driver of a mother's preferences: updating beyond the firstborn holds for all three outcomes in the pooled interaction (Table A.2), though the total-children result is noisy within individual sibship cells.

4.3 Heterogeneity

A large literature documents that a mother's fertility preferences depend on her circumstances, including her age and urban residence (Beguy and Mberu, 2015), household finances (Black et al., 2013; Sennott and Yeatman, 2012), education (Caldwell, 1980), and experience of child mortality (Shapiro and Gebreselassie, 2008), as well as relationship status (Sennott and Yeatman, 2012), age at first marriage (Adhikari, 2010; Garenne, 2008), labor force participation (Garenne, 2008), health (Alemayehu and Berhe, 2012; Finocchario-Kessler et al., 2010), religion (Adhikari, 2010), and contraceptive use (Garenne, 2008). If preferences depend on these circumstances, a mother's response to a recent birth may as well. In this section, I test whether the treatment effect varies with the characteristics observed in my data: rural residence, poverty, age, age at first birth, education, and the death of a child.

4.3.1 Country-level SRBs

Section 3.2 addresses whether sex-selective abortion threatens identification at the level of the pooled sample. Here I test a complementary question: does the treatment effect itself vary with a country's sex ratio at birth? This would be a further warning sign of selective behavior concentrated in high-SRB settings such as Afghanistan (119.3 males per 100 females, well above the natural range).

To investigate this issue, I rely on an OLS specification similar to my main specification, but with separate samples for country SRBs by quintile. Results of this analysis are in Tables A.6–A.8. The treatment effect on ideal daughters is nearly identical across quintiles (0.114 in Q1 vs. 0.108 in Q5, both significant beyond the 1% level), with no monotonic gradient across Q1–Q5 (Table A.6). The effect on ideal sons, by contrast, attenuates noticeably in magnitude in the extremes (both Q1 and Q5) (Table A.7), showing that mothers from countries with extreme attitudes on the ideal sibship's sex composition may have stickier preferences specifically with regard to sons. Combined with the identification diagnostics in Section 3.2, this provides further evidence that treatment effects, especially on daughters, are not being driven by systematic distortion of the SRBs.

4.3.2 Individual Characteristics

I test for heterogeneous treatment effects across six of my included control variables, each of which the literature has previously established as influential on a mother's preferences. For the continuous control variables, I have created dummy variables for mothers with above-average values of in my sample (e.g., above average age). The OLS speci-

fications used to test for heterogeneous treatment effects are the same as my primary OLS specification except for the inclusion of the interaction term between the treatment and the individual characteristic of choice.

Tables A.9–A.11 display the results of these specifications. Table A.9 shows that there are no significant heterogeneous treatment effects for any individual characteristics when the ideal number of daughters is the dependent variable. Table A.10 shows that rural mothers experience weaker treatment effects than urban mothers when the ideal number of sons is the dependent variable, and this is significant at the 1% level; however, this is one of only two tests (of 36 run across this section and Section 4.3.3 combined) that survives a Benjamini-Hochberg correction for multiple testing, and only at a 10% false discovery rate (Table A.15). There are no heterogeneous treatment effects besides those of rural and urban mothers. Table A.11 shows that there are no statistically significant heterogeneous treatment effects for the chosen individual characteristics when the ideal number of total children is the dependent variable. The homogeneity of treatment effects across groups with different individual characteristics is striking. Having experienced the death of a child does not moderate the treatment effect for any of the three outcomes, so replacement-fertility motives or the most brutal sex-selective practices do not appear to be driving the results.

Likewise, there is no strong evidence that mothers of different ages experience statistically different treatment effects after a birth. Figure A.3 shows the treatment effect estimated separately by respondent age group. The coefficient on ideal daughters is positive across nearly the entire 15–38 age range, generally between 0.05 and 0.20 and statistically significant for most bins between ages 22 and 33; there is no evidence that preference updating is concentrated among younger or older mothers specifically.

Across these various individual characteristics, heterogeneous treatment effects are remarkably absent despite the number of tests. This speaks to the strength and salience of these results; on average, mothers from very different socioeconomic backgrounds seem to update their preferences with striking similarity.

4.3.3 Cultural and Macroeconomic Moderators

Thus far, no subsample has yielded directionally heterogeneous treatment effects on ideal daughters or sons relative to the primary OLS specifications. That said, there is some wavering of treatment effects at country-SRB extremes, so perhaps there is some other moderator of treatment effects. To this end, I search for stronger moderators of treatment effects by examining pertinent macroeconomic and cultural indicators. Chosen potential moderators are at the country level. These include ln(GDPpc), Gender Equality Rating, Gini Index, Human Capital Index (HCI), Direction of Dowry, and Country TV Penetration. All of these variables have been associated with differing individual preferences.[6]

The specifications used in Tables A.12–A.14 are similar to those used to comb for possible heterogeneous treatment effects from individual characteristics. Of course, the macroeconomic/cultural indicator is used in place of the individual characteristic, and the interaction term is the treatment variable recent daughter birth interacted with the given macroeconomic/cultural indicator. Fixed effects are also removed to avoid collinearity issues as these indicators are at the country level.

Table A.12 speaks to heterogeneous treatment effects of these macroeconomic/cultural indicators when the ideal number of daughters is the dependent variable. The effect on ideal daughters is positive in every group, so treatment effects

6 ln(GDPpc) has been associated with economic preferences such as patience (Falk et al., 2018). Gender inequality has been shown to shape women's preferences (Falk and Hermle, 2018). Economic inequality, captured here by the Gini Index, has been shown to alter individuals' preferences for certain goods (Velandia-Morales, Rodríguez-Bailón and Martínez, 2022). Elements of human capital such as literacy (Adhikari, 2010), education (Caldwell, 1980), and health (Alemayehu and Berhe, 2012; Finocchario-Kessler et al., 2010) have been found to influence fertility preferences, and an individual's human capital influences the human capital of their children (Qin, Wang and Zhuang, 2016), so by extension it is likely that human capital at a country level may influence fertility preferences as well. Dowry practices shape fertility preferences (Diamond-Smith, Luke and McGarvey, 2008). Media exposure has also been linked to shifts in sex-related preferences and fertility behavior (e.g., cable television and women's status in India; telenovelas and fertility decline in Brazil) (Jensen and Oster, 2009; La Ferrara, Chong and Duryea, 2012), motivating country TV penetration as a candidate moderator.

are directionally homogeneous. Mothers in high ln(GDPpc) countries show a somewhat larger increase (interaction of 0.032, $p < 0.10$), and mothers in high-TV-penetration countries show a larger one (0.039, $p < 0.05$). The interactions with the Gini index and HCI are small and statistically insignificant, as are those with gender equality and the direction of dowry. Neither the ln(GDPpc) nor the TV interaction survives the multiple-testing correction in Table A.15.

Table A.13 shows identical specifications to Table A.12, but where the ideal number of sons is the dependent variable instead. Again, gender equality rating displays no heterogeneous treatment effects. Heterogeneity is more apparent here than for ideal daughters, though it is concentrated in a few moderators. Mothers in countries with above-average HCI show a significantly larger reduction in ideal sons after a daughter birth (interaction of -0.042, $p < 0.05$), as do mothers in high-TV-penetration countries (-0.034, $p < 0.05$). The interactions with ln(GDPpc) and the Gini index are negative but not statistically significant. Interestingly, mothers in countries where the dowry historically goes to the groom's family exhibit especially heterogeneous treatment effects. The ideal number of sons desired for mothers in countries where this dowry practice has not historically existed decreases by 0.105 after the birth of a daughter. But in countries where the dowry goes to the groom, this treatment effect increases by 0.056, and this is significant at the 1% level. Treatment effects in countries with different dowry practices are very heterogeneous: the effect on ideal sons is roughly halved where the dowry goes to the groom. Preferences towards sons appear to be far stickier in countries where the dowry has historically gone to the groom's family.

Table A.14 displays heterogeneous treatment effects for specifications with ideal number of total children as the outcome variable. Treatment effects are homogeneous across high and low levels of ln(GDPpc), gender equality rating, Gini Index, and HCI (at the 5% level). That said, estimates here are unreliable as the coefficients on recent daughter birth here in these aforementioned specifications are spotty in significance. Mothers in countries with dowry-to-groom practices have significant and positive treatment effects, but mothers in countries without these practices do not have significant treatment effects. Not much can be inferred from Table A.14 due to minimal statistical significance across the board.

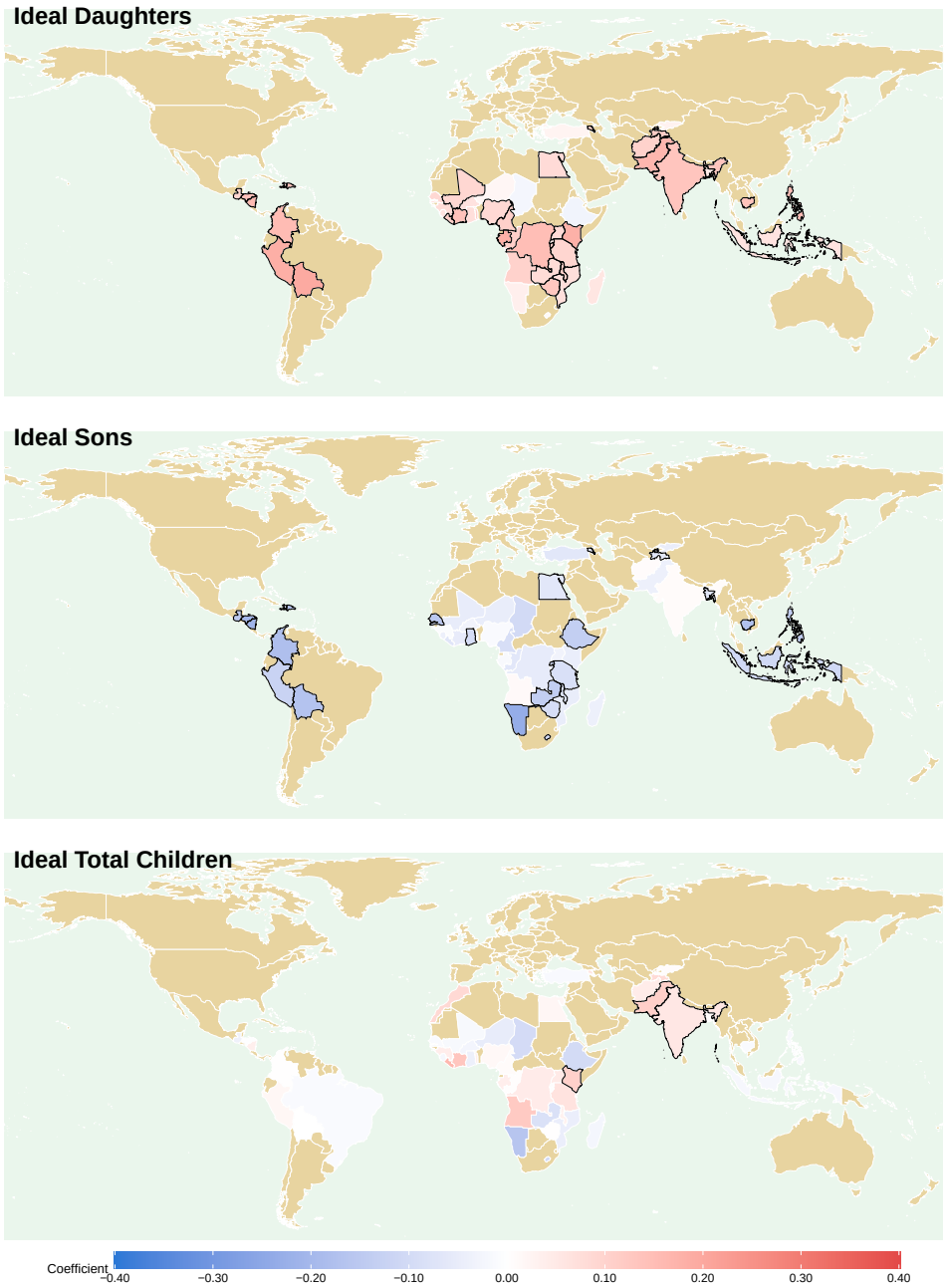
Zooming out even further, Figure A.2 plots the country-specific OLS coefficients for each outcome, and Figure 4 maps these estimates geographically using the same color scale across all three outcomes. Of the 56 countries with a sufficient sample size, 34 have a statistically significant positive coefficient on ideal daughters and 28 have a statistically significant negative coefficient on ideal sons; not a single country shows a significant coefficient of the opposite sign for either outcome.

Unlike individual characteristics, there does seem to be heterogeneity in treatment effects of mothers from different cultures under robust standard errors. Relatively rich countries and countries with high TV penetration seem to exhibit stronger treatment effects on the ideal number of girls, while countries where the dowry is given to the groom's family exhibit stickier son preferences than other countries. However, once standard errors are clustered by country and corrected for multiple testing, only the dowry-to-groom$\times$sons interaction survives, and only at a 10% false discovery rate (Table A.15).

4.3.4 Year of Survey

Another concern is that treatment effects changed strongly over the years, given the progressive diffusion of sex-selective technologies in some of the sampled populations over recent years. Reassuringly, survey year does not seem to strongly affect treatment effects for any specification. Figure A.4 shows specifications estimated separately by survey year rather than by country or respondent age. The coefficient on ideal daughters ranges from about 0.04 to 0.16 across survey years from 1994 to 2022, with no discernible time trend.

Figure 4: Country-Level Treatment Effects Visualized



Notes: Each map shows the country-level treatment effect on one outcome (top to bottom: ideal daughters, ideal sons, ideal total children), with countries labeled by panel tag rather than a repeated title. Blue–white–red diverging color scale centered at zero: blue always denotes a negative coefficient and red always denotes a positive coefficient, applied identically in every panel. All three panels share the same color scale, so magnitudes are directly comparable across outcomes. Black border = statistically significant at the 5% level. Grey = country not in the analysis sample or with fewer than 1,000 usable observations; grey is deliberately darker than the near-white shade used for a near-zero coefficient so the two are not visually confused.

4.4 Floor Tests

One possible explanation for the observed results is that they may reflect a purely mechanical artifact, where mothers never report an ideal number of sons, daughters, or total children below what they already have. Encouragingly, Figure A.5 shows that the floor does not bind strictly, though relatively few mothers report an ideal below their current count.

To further investigate this possibility, I restrict the sample to two groups where floor censoring cannot explain the treatment effect. First, I include only mothers who desire at least 2 more daughters (sons, total children) than they currently have. Because a single birth shifts the ideal-minus-actual gap by exactly one unit, these women remain above the floor regardless of the sex of the recent birth; the censoring constraint cannot mechanically raise their stated ideal. Second, I include only mothers who already have more daughters/sons/total children than they ideally desire. These women have already overshot their stated preference, so no upward floor pressure applies from any direction. A significant treatment effect in either subsample is evidence of preference updating that cannot be attributed to floor censoring. Because the gap used to define these strata is itself a function of the stated, post-birth ideal, this exercise is best understood as a bounding/diagnostic check on the floor-artifact story rather than a clean causal comparison across subgroups: treatment can shift mothers across strata, so the resulting within-stratum coefficients should not be read as the causal effect specific to that subgroup.

Table A.16 shows these results. In the Gap ≥ 2 stratum, women already state an ideal at least 2 above their actual count, so floor censoring cannot mechanically bind. Here, the treatment effect is 0.8297 on ideal daughters and -0.8062 on ideal sons. Both are significant beyond the 1% level. The effect on ideal total children is 0.0146 and is not significant. In the Gap ≤ -1 stratum, women already state an ideal below their actual count, and the treatment effect is 0.5363 on ideal daughters and -0.5812 on ideal sons. Both are significant beyond the 1% level. The effect on ideal total children is -0.0108 and is not significant. In both of these gap designs, the results for ideal daughters and sons survive and remain significant, ruling out a purely mechanical floor-censoring explanation for the primary results. The point estimates within these strata are larger in magnitude, but this should not be read as evidence that mothers farther from their stated ideal are more responsive to treatment: stratifying on the stated gap conditions on a post-treatment variable, so these magnitudes are reported only as a diagnostic bound on the floor-artifact story, not as a causal subgroup comparison.

5 Mechanisms

Two primary mechanisms are likely candidates to be driving the results: contact and ex-post rationalization. In this section, I show that the evidence is more consistent with the expectations of ex-post rationalization than contact.

If contact is driving the observed effects, we would expect the treatment effects to grow as time passes after the birth, as mothers are more and more exposed to the experience of having a daughter/son/more children. Conversely, if mothers are rationalizing their outcomes ex-post, we would expect the treatment effects to emerge directly after birth and be stable with time. To test for instability of a mother's preferences in the months following a birth, I interact the treatment (recent daughter birth) with months since the most recent birth as a continuous variable (0–12 months). Table 4 shows the results of this specification for all three outcomes. The interaction term is small and statistically insignificant for all three outcomes (0.0006 for ideal daughters, 0.0014 for ideal sons, -0.0017 for ideal total children), indicating the treatment effect neither grows nor fades over the year following the birth. Figure A.6 presents a complementary month-by-month exposure gradient. The month-by-month coefficient on ideal daughters is roughly flat across the 0–12 month window (fluctuating between about 0.04 and 0.16, with a linear trend that is nearly indistinguishable from flat).

A second test of contact exploits births to other women in the mother's household. If exposure to a daughter shifts preferences, mothers who live with a newborn girl but did not give birth to her should also update, if more weakly. I

Table 4: Timing Test: Continuous Interaction with Months Since Birth

	(1) Ideal Daughters	(2) Ideal Sons	(3) Ideal Total Children
Recent Daughter Birth	0.1064***	-0.0652***	0.0462**
	(0.0132)	(0.0156)	(0.0180)
Recent Daughter Birth x Months Since Birth	0.0006	0.0014	-0.0017
	(0.0019)	(0.0022)	(0.0026)
Observations	239,450	239,449	282,824
R-squared	0.3477	0.3538	0.5157

Notes: Interacts the treatment (recent daughter birth) with months since the most recent birth as a continuous variable (0–12 months), for each of the three outcomes (ideal daughters, sons, and total children, one column each). The coefficient on the interaction term tests whether the treatment effect grows over time. Specification: full sibship controls; region×interview-month and interview-year FE; population-weighted; robust SEs. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

restrict to households with exactly one recent birth (within 12 months) that is a singleton, alive, co-resident with its mother, and her first to third birth, mirroring the main design. Focal mothers are other women in the household with one to three children of their own and no recent birth. I drop pairs in which the newborn is the focal mother's grandchild or sibling, pairs with an unknown relationship to the household head, and surveys in which household identifiers are not unique. The treatment is whether the co-resident's recent birth is a daughter. I control for the focal mother's own sibship composition, the standard covariates, the number of women interviewed in the household, other women's living sons and daughters, and the co-resident's parity and months since birth, with region-by-interview-month, interview-year, and relationship-type fixed effects. Table 5 shows the results for 18,951 focal mothers. The sex of a co-resident's recent birth has no detectable effect on any outcome (-0.016 for ideal daughters, 0.012 for ideal sons, -0.047 for ideal total children). Daily exposure to a newborn daughter in the household does not shift preferences when the child is not the mother's own, though contact with a co-resident's child is plausibly less intense than with one's own.

Table 5: Household Contact: Co-resident's Recent Birth

	Ideal Daughters		Ideal Sons		Ideal Total Children	
	(1) Parsim.	(2) Full	(3) Parsim.	(4) Full	(5) Parsim.	(6) Full
Co-resident Recent Daughter Birth	-0.0141	-0.0156	0.0196	0.0123	-0.0318	-0.0474
	(0.0332)	(0.0328)	(0.0376)	(0.0371)	(0.0534)	(0.0522)
Observations	19,211	18,951	19,211	18,951	19,211	18,951
R-squared	0.4172	0.4302	0.4100	0.4208	0.5853	0.6037

Notes: Sample: mothers with 1–3 children and no recent birth of their own who live in a household with exactly one recent birth ($\leq$12 months) to another woman; the birth is a singleton, alive, co-resident with its mother, and her first to third birth. Pairs in which the newborn is the focal mother's grandchild or (half-)sibling, pairs with an unknown relationship to the household head, and surveys with non-unique household identifiers are excluded. Treatment: the co-resident's recent birth is a daughter. "Parsim." columns include no controls; "Full" columns control for the focal mother's sibship composition dummies, age, age at first birth, education (years), rural, poor, marital status (unmarried/widowed/divorced, married omitted), number of women interviewed in the household, other women's living sons and daughters, and the co-resident's parity and months since birth. All columns absorb region×interview-month, interview-year, and relationship-type fixed effects; population-weighted; SEs clustered by survey cluster. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

These results are more consistent with ex-post rationalization than with contact. Completely ruling out either mechanism, however, would likely require examining mothers' preferences prior to the reveal of the sex of their child and examining mothers' preferences over longer time horizons, but successive births make isolating effects challenging.

6 Discussion

Relative to mothers with recent son births, mothers with recent daughter births desire a higher ideal number of daughters, a lower ideal number of sons, and a higher ideal number of total children. The first two of these results are consistent across a breadth of specifications, hovering around 0.11 for ideal number of daughters and 0.05 for ideal number of sons; treatment effect estimates for these two outcomes are remarkably robust and entirely directionally homogeneous. The result for ideal total children, by contrast, is less robust: the point estimate hovers around 0.04, but this is accompanied by insignificant estimates in some subsamples, only 2 of 7 significant sibship-cell specifications, and a PSM average treatment effect on the treated that is small, insignificant, and of the opposite sign (Table A.18). I therefore treat the composition results (daughters, sons) as the paper's robust headline finding, and the level result (total children) as suggestive but fragile.

This paper takes initial steps toward testing the mechanisms behind these effects. The shift in preferences appears immediately after birth, remains stable over the following year, and does not extend to mothers living with another woman's newborn. This pattern is more consistent with mothers reconciling their preferences with an outcome they did not choose than with preferences that evolve through accumulated contact. The absence of signatures of contact effects is notable given that intergroup contact has been shown to reduce prejudice in India, a setting where son preference is pronounced (Rao, 2019; Lowe, 2021): contact may well operate in these settings, but it does not appear to drive the shift in stated preferences within the first year after birth. Natural experiments at the scale afforded by DHS data are a useful complement to laboratory evidence drawn overwhelmingly from Western, Educated, Industrialized, Rich, and Democratic (WEIRD) populations (Henrich, Heine and Norenzayan, 2010; Norenzayan and Heine, 2005), and the results provide a useful, if partial, piece of evidence that ex-post rationalization is not unique to WEIRD contexts. More humbly, this paper substantiates a psychological regularity first outlined in Lamson (2020): across different individual characteristics and macroeconomic and cultural indicators, mothers with a recent daughter birth tend to prefer a greater ideal number of daughters and a lower ideal number of sons than those who recently birthed a son.

There are almost no heterogeneous treatment effects across groups possessing high and low levels of different individual characteristics known to influence fertility preferences or across macroeconomic and cultural indicators. Once standard errors are clustered by country and corrected for the 36 heterogeneity tests run across Sections 4.3.2 and 4.3.3 (Table A.15), only two interactions survive at a 10% false discovery rate—rural residence$\times$sons among individual characteristics and dowry-to-groom$\times$sons among macroeconomic and cultural indicators—and none survive at 5%. Mothers in countries where the dowry has historically gone to the groom's family do still show stickier, smaller-magnitude son effects, consistent with a simple "son-preference environments dampen updating" story—but the associations with $\ln(\text{GDPpc})$, Gini, and HCI reported under robust standard errors do not hold up under this stricter standard. The directional homogeneity and near-total absence of heterogeneity that survives scrutiny merit thinking about the meaning of these treatment effects on a deeper level.

Past research has found that the sex of the eldest child exhibits particularly strong effects on fertility (Dahl and Moretti, 2008; Jayachandran and Pande, 2017; Tavassoli, 2021). This paper shows that preference updating is not confined to the firstborn for ideal sons and ideal total children: mothers continue to update these preferences at later births, and the firstborn does not appear to be the primary driver. For ideal daughters, the firstborn interaction in Table A.2 is instead significantly negative, meaning updating is somewhat *stronger*, not weaker, beyond the firstborn—if anything, the opposite of a firstborn-dominated pattern. This finding supports conclusions of Müller et al. (2022), which indicates that mothers update their fertility preferences with each birth.

This paper informs potential policy targeting and measurement concerns. If stated preferences on ideal sibship compositions are themselves responsive to the sex of a recent birth, then survey-based measures of son preference used

to target or evaluate interventions (e.g., information campaigns against sex-selective abortion) may mismeasure son preference among households based simply on the timing and sex of their most recent birth, rather than on any stable difference in underlying preference. The same complication extends to the measurement of unwanted fertility and cross-population measurement more broadly: approaches that treat stated ideal fertility and sex composition as a fixed structural input, whether to estimate unwanted fertility or to benchmark the intensity of son preference across countries and over time, implicitly assume these preferences are exogenous to realized births. The results here suggest otherwise: stated preferences are themselves partly an output of recent fertility history, so a cross-sectional snapshot of "how much son preference exists" in a given population may partly reflect the composition of recent births in that sample rather than only a stable underlying parameter.

Chief among the limitations of this paper is my lack of panel data, as I cannot track mothers' preferences over time as done in Müller et al. (2022), or compare preferences directly pre- and post-birth. The necessary restriction to only mothers of recent births does not allow me to observe the true effect of a recent daughter birth in isolation, only the influence of a recent daughter birth relative to the influence of a son. An additional limitation is external validity. The 82 countries in this sample are drawn exclusively from the DHS, a survey program fielded in low- and middle-income countries across Sub-Saharan Africa, Asia, Latin America, Central Asia, and MENA. The results here therefore speak to preference updating in the developing-country contexts where son preference and high fertility are most consequential for the global sex ratio, which is precisely the population of greatest policy relevance. That said, they should not be assumed to generalize to low-fertility, high-income settings where the underlying mechanisms may operate against a very different baseline of family-size norms and sex-selection technology.

While mothers have little control over the sex composition of their sibship without resorting to extreme measures, they often adjust their preferences towards the sex of the child to which they have given birth. In turn, these shifting preferences may influence fertility outcomes, which themselves shape future preferences. Future research should involve panel data to validate these treatment effects by tracking individual preferences over long time horizons both pre- and post-birth of a child, and further investigate the possible role of ex-post rationalization in driving the observed effects.

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A Appendix

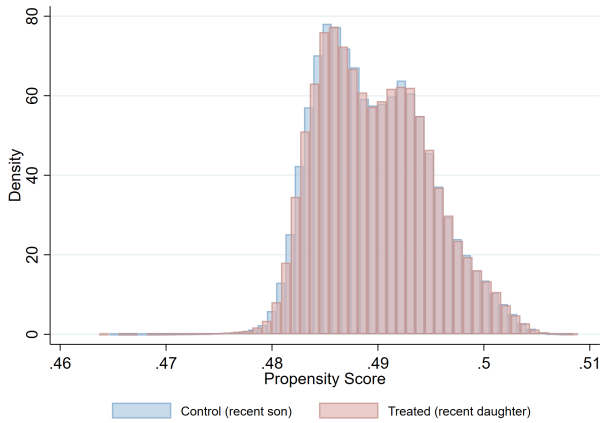
A.1 Identification Diagnostics

Table A.1: SRB Orthogonality: Recent Birth Sex on Prior Family Composition

	(1) Recent Daughter Birth
Sibship: Son	-0.0002 (0.0065)
Sibship: Daughter	0.0040 (0.0064)
Sibship: Son-Son	0.0111 (0.0093)
Sibship: Son-Daughter	0.0139 (0.0095)
Sibship: Daughter-Son	-0.0054 (0.0094)
Sibship: Daughter-Daughter	-0.0029 (0.0095)
Constant	0.4844*** (0.0035)
Observations	306,294
R-squared	0.0005

Notes: Dependent variable: indicator for whether the most recent birth was a daughter. Regressors: indicators for each pre-existing family-composition cell. Coefficients near zero indicate that the sex of the most recent birth is orthogonal to prior family composition, consistent with quasi-random assignment. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure A.1: Common Support



Notes: Kernel density plots of the estimated propensity score (predicted probability of a recent daughter birth, from a logit on age, age at first birth, education years, rural, and poor) for treated (recent daughter, pink) and control (recent son, blue) groups. Substantial overlap across the two distributions confirms common support and validates the PSM approach: treated and control mothers are observationally comparable, consistent with the quasi-randomness of birth sex.

A.2 Updating Beyond the Firstborn

Table A.2: Firstborn Interaction

	(1) Ideal Daughters	(2) Ideal Sons	(3) Ideal Total Children
Recent Daughter Birth	0.1220***	-0.0603***	0.0411***
	(0.0090)	(0.0103)	(0.0125)
Child is Firstborn	-0.1000***	-0.1366***	-0.2215***
	(0.0118)	(0.0139)	(0.0165)
Recent Daughter Birth x Firstborn	-0.0301**	0.0122	-0.0100
	(0.0136)	(0.0159)	(0.0184)
Constant	1.4132***	1.7272***	3.3613***
	(0.0237)	(0.0274)	(0.0338)
Observations	239,578	239,577	282,952
R-squared	0.3429	0.3511	0.5122

Notes: Augments the baseline sibship specification with an indicator for no pre-existing sibship (i.e., the recent birth is the respondent's first child) interacted with recent daughter birth. The main effect on recent daughter birth is the treatment effect for mothers with a pre-existing sibship; the interaction coefficient tests whether this treatment effect differs for mothers whose recent birth is their first child. Specification: full sibship controls plus interaction; region×interview-month and interview-year FE; population-weighted; robust SEs. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.3: OLS by Sibship Cell — Ideal Daughters

	(1) None	(2) S	(3) D	(4) SS	(5) SD	(6) DS	(7) DD
Recent Daughter Birth	0.0911***	0.1383***	0.1092***	0.0809***	0.1755***	0.1742***	0.0600*
	(0.0103)	(0.0153)	(0.0165)	(0.0300)	(0.0332)	(0.0337)	(0.0361)
Constant	1.2429***	1.3312***	1.5041***	1.6196***	1.5908***	1.6909***	1.8970***
	(0.0348)	(0.0507)	(0.0585)	(0.1127)	(0.1192)	(0.1194)	(0.1252)
Observations	96,772	41,069	39,202	14,584	13,837	13,515	13,740
R-squared	0.3825	0.4054	0.3989	0.4422	0.4297	0.4584	0.4257

Notes: Each column runs the baseline OLS specification separately within a specific pre-existing sibling composition (sibship cell). S = prior son, D = prior daughter; the label denotes the sequence of births up to the penultimate child (e.g., SD = firstborn son, secondborn daughter). The identifying comparison within each cell is between women whose most recent birth was a daughter vs. a son, conditional on identical prior sibship composition. Cells with too few observations are omitted. Specification: age, age at first birth, education, rural, poor; region×interview-month and interview-year FE; population-weighted; robust SEs. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.4: OLS by Sibship Cell — Ideal Sons

	(1) None	(2) S	(3) D	(4) SS	(5) SD	(6) DS	(7) DD
Recent Daughter Birth	-0.0474***	-0.0430**	-0.0725***	-0.0869**	-0.0446	-0.0428	-0.0254
	(0.0123)	(0.0193)	(0.0189)	(0.0394)	(0.0363)	(0.0418)	(0.0335)
Constant	1.4497***	1.7802***	1.8328***	2.2153***	1.9032***	2.0094***	1.8168***
	(0.0397)	(0.0640)	(0.0627)	(0.1481)	(0.1281)	(0.1563)	(0.1245)
Observations	96,772	41,069	39,201	14,584	13,837	13,515	13,740
R-squared	0.3940	0.3822	0.4314	0.3965	0.4124	0.4309	0.4543

Notes: See notes to Table A.3.

Table A.5: OLS by Sibship Cell — Ideal Total Children

	(1) None	(2) S	(3) D	(4) SS	(5) SD	(6) DS	(7) DD
Recent Daughter Birth	0.0300**	0.0097	0.0772***	-0.0347	0.0860*	0.0590	0.0559
	(0.0135)	(0.0208)	(0.0219)	(0.0392)	(0.0458)	(0.0426)	(0.0485)
Constant	3.0836***	3.4192***	3.3976***	3.8942***	3.7686***	4.0017***	3.8093***
	(0.0453)	(0.0834)	(0.0748)	(0.1410)	(0.1747)	(0.1504)	(0.1572)
Observations	114,294	48,500	46,321	17,351	16,466	16,175	16,414
R-squared	0.5295	0.5474	0.5567	0.5964	0.5725	0.5994	0.5758

Notes: See notes to Table A.3.

A.3 High and Low SRB Samples

Table A.6: Treatment Effect by Country SRB Quintile — Ideal Daughters

	(1) Q1 (Lowest)	(2) Q2	(3) Q3	(4) Q4	(5) Q5 (Highest)
Recent Daughter Birth	0.1140***	0.1096***	0.1155***	0.0992***	0.1077***
	(0.0250)	(0.0348)	(0.0151)	(0.0126)	(0.0099)
Observations	47,781	18,481	56,168	52,021	45,532
R-squared	0.1846	0.2460	0.3441	0.3409	0.2897

Notes: Each column restricts the sample to countries within a quintile of the country-level sex ratio at birth (SRB), computed as the ratio of male to female births in the analysis sample. Q1 = lowest-SRB countries (≈103 males per 100 females); Q5 = highest-SRB countries (includes countries with anomalous SRB consistent with sex-selective practices). Within each quintile the treatment effect of a recent daughter birth on stated ideal daughters is estimated with region×interview-month and interview-year fixed effects, population weights, and robust SEs. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.7: Treatment Effect by Country SRB Quintile — Ideal Sons

	(1) Q1 (Lowest)	(2) Q2	(3) Q3	(4) Q4	(5) Q5 (Highest)
Recent Daughter Birth	-0.0581**	-0.0805**	-0.1036***	-0.0695***	-0.0377***
	(0.0266)	(0.0359)	(0.0152)	(0.0144)	(0.0122)
Observations	47,781	18,481	56,168	52,020	45,532
R-squared	0.2084	0.2714	0.4103	0.3649	0.3523

Notes: Dependent variable: stated ideal sons. See notes to Table A.6. In high-SRB countries the son-reducing effect of a daughter birth may be attenuated if son preferences are more rigid or if the daughter birth is followed by continued son-seeking behavior.

Table A.8: Treatment Effect by Country SRB Quintile — Ideal Total Children

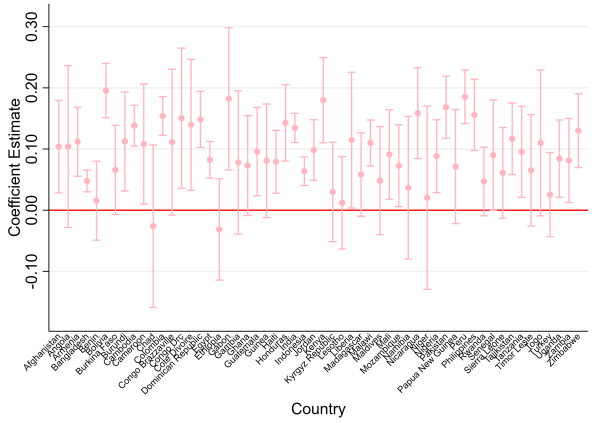
	(1) Q1 (Lowest)	(2) Q2	(3) Q3	(4) Q4	(5) Q5 (Highest)
Recent Daughter Birth	0.0660	0.0792	-0.0098	0.0346*	0.0379***
	(0.0417)	(0.0539)	(0.0234)	(0.0193)	(0.0115)
Observations	53,837	21,211	67,571	58,042	59,766
R-squared	0.4508	0.4051	0.6199	0.5691	0.3706

Notes: Dependent variable: stated ideal total children. See notes to Table A.6.

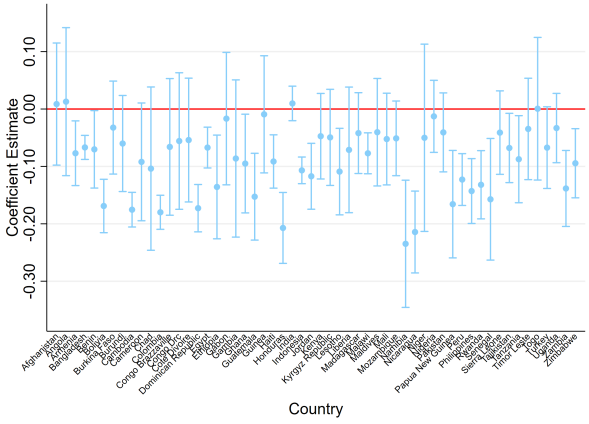
A.4 By-Country Results

Figure A.2: By-Country Estimates

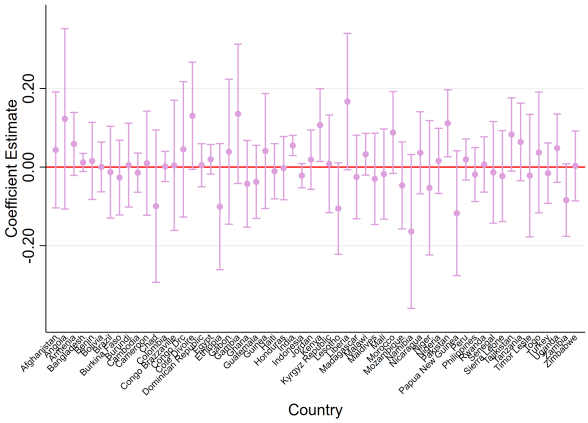
(a) Ideal Daughters



(b) Ideal Sons



(c) Ideal Total Children



Notes: Country-level OLS coefficient of recent daughter birth on stated ideal daughters, sons, and total children (Panels (a)–(c)), estimated separately for each of the 82 DHS countries. Each point = one country; capped lines = 95% CIs. Countries are sorted by point estimate. Specification within each country: sibship dummies, age, age at first birth, education, rural, poor, marital status (unmarried/widowed/divorced, married omitted), age at first marriage, and child mortality count; region×interview-month and interview-year FE; population-weighted; robust SEs. Countries with fewer than 1,000 usable observations are excluded.

A.5 Near-universal Directional Homogeneity

Table A.9: Individual Heterogeneity — Ideal Daughters

	(1) Rural	(2) Poor	(3) Abv-Avg Age	(4) Abv-Avg Age FB	(5) Abv-Avg Educ	(6) Child Death
Recent Daughter Birth	0.1098***	0.1149***	0.1033***	0.1087***	0.1157***	0.1117***
	(0.0106)	(0.0080)	(0.0089)	(0.0090)	(0.0100)	(0.0071)
Individual Characteristic	0.0852***	0.0486***	0.0169	-0.0038	0.0012	0.0222
	(0.0108)	(0.0130)	(0.0132)	(0.0129)	(0.0144)	(0.0162)
Interaction Term	0.0010	-0.0151	0.0177	0.0044	-0.0117	-0.0116
	(0.0137)	(0.0152)	(0.0139)	(0.0137)	(0.0135)	(0.0234)
Constant	1.3269***	1.3245***	1.3695***	1.3250***	1.3242***	1.3260***
	(0.0227)	(0.0225)	(0.0282)	(0.0283)	(0.0226)	(0.0223)
Observations	239,578	239,578	239,578	239,578	239,578	239,578
R-squared	0.3414	0.3414	0.3415	0.3414	0.3414	0.3414

Notes: Each column interacts treatment with a binary individual characteristic (rural, poor, above-average age, above-average age at first birth, above-average education, having experienced the death of a child). Region×month and year FE; population-weighted; robust SEs. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.10: Individual Heterogeneity — Ideal Sons

	(1) Rural	(2) Poor	(3) Abv-Avg Age	(4) Abv-Avg Age FB	(5) Abv-Avg Educ	(6) Child Death
Recent Daughter Birth	-0.0877***	-0.0498***	-0.0472***	-0.0520***	-0.0435***	-0.0589***
	(0.0121)	(0.0095)	(0.0106)	(0.0106)	(0.0118)	(0.0082)
Individual Characteristic	0.0931***	0.0178	0.0355**	-0.0152	0.0327**	0.0281
	(0.0129)	(0.0146)	(0.0154)	(0.0150)	(0.0163)	(0.0188)
Interaction Term	0.0502***	-0.0153	-0.0175	-0.0058	-0.0238	0.0409
	(0.0158)	(0.0171)	(0.0158)	(0.0157)	(0.0154)	(0.0275)
Constant	1.6318***	1.6138***	1.6539***	1.5867***	1.6102***	1.6182***
	(0.0262)	(0.0260)	(0.0325)	(0.0326)	(0.0263)	(0.0258)
Observations	239,577	239,577	239,577	239,577	239,577	239,577
R-squared	0.3499	0.3498	0.3498	0.3498	0.3498	0.3498

Notes: See notes to Table A.9.

Table A.11: Individual Heterogeneity — Ideal Total Children

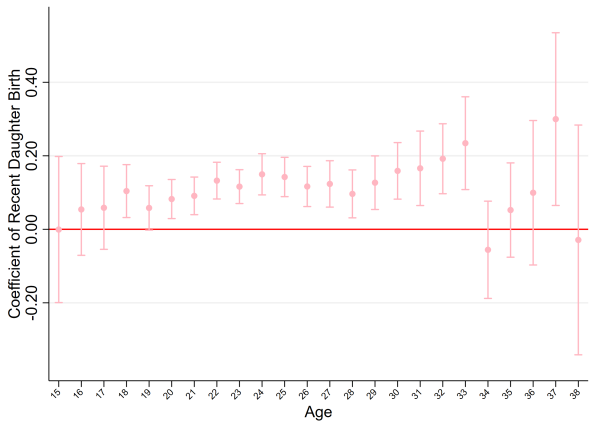
	(1) Rural	(2) Poor	(3) Abv-Avg Age	(4) Abv-Avg Age FB	(5) Abv-Avg Educ	(6) Child Death
Recent Daughter Birth	0.0248	0.0440***	0.0375***	0.0374***	0.0443***	0.0370***
	(0.0161)	(0.0104)	(0.0122)	(0.0124)	(0.0132)	(0.0096)
Individual Characteristic	0.1690***	0.1569***	0.0459**	-0.0133	0.0019	0.1180***
	(0.0152)	(0.0201)	(0.0183)	(0.0179)	(0.0205)	(0.0227)
Interaction Term	0.0219	-0.0276	0.0038	0.0042	-0.0132	0.0170
	(0.0196)	(0.0228)	(0.0188)	(0.0186)	(0.0179)	(0.0334)
Constant	3.1751***	3.1660***	3.2440***	3.1514***	3.1658***	3.1693***
	(0.0322)	(0.0315)	(0.0403)	(0.0394)	(0.0316)	(0.0315)
Observations	282,952	282,952	282,952	282,952	282,952	282,952
R-squared	0.5107	0.5107	0.5107	0.5107	0.5107	0.5107

Notes: See notes to Table A.9.

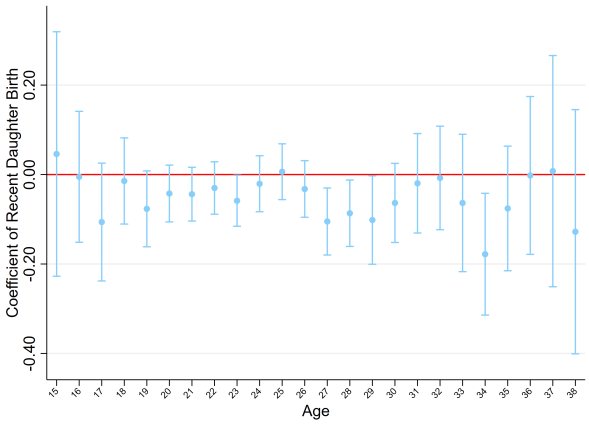
A.6 By-Age Results

Figure A.3: By-Age Coefficients

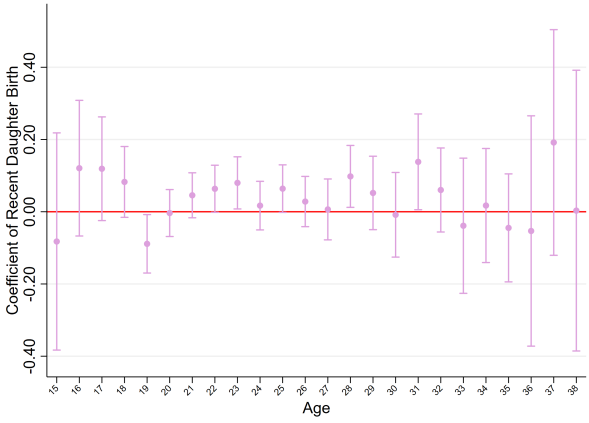
(a) Ideal Daughters



(b) Ideal Sons



(c) Ideal Total Children

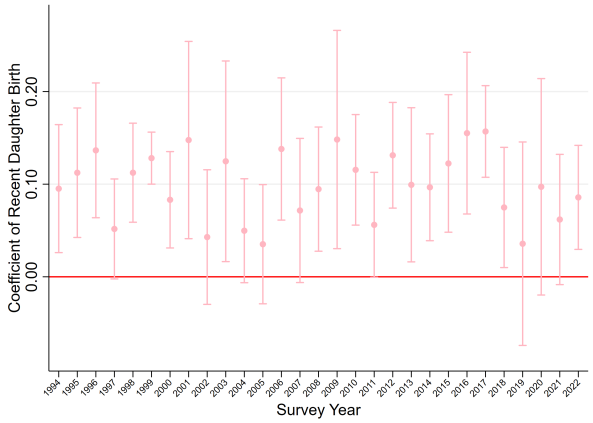


Notes: OLS coefficient of recent daughter birth on stated ideal daughters, sons, and total children (Panels (a)–(c)), estimated separately for each respondent age group. Each point = one age bin; capped lines = 95% CIs. Age bins with fewer than 200 observations are excluded. Specification: sibship dummies, age at first birth, education, rural, poor, marital status (unmarried/widowed/divorced, married omitted), age at first marriage, and child mortality count; region×interview-month and interview-year FE; population-weighted; robust SEs. (Age itself is part of the shared control set but is collinear-dropped within each single-age bin by construction.) Variation across age groups reveals whether preference updating is concentrated among younger mothers (who have more remaining fertility) or is broadly distributed.

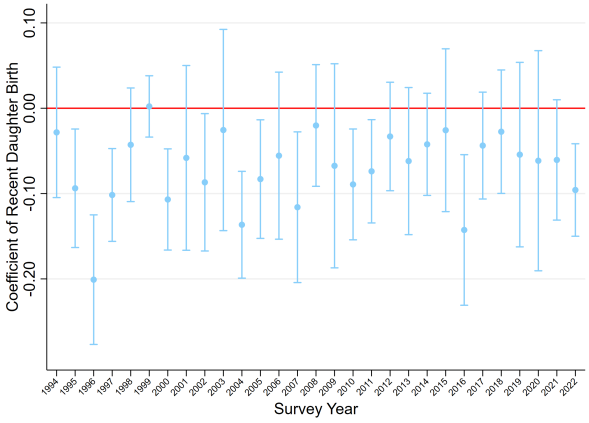
A.7 By-Year-of-Survey Results

Figure A.4: By-Survey-Year Coefficients

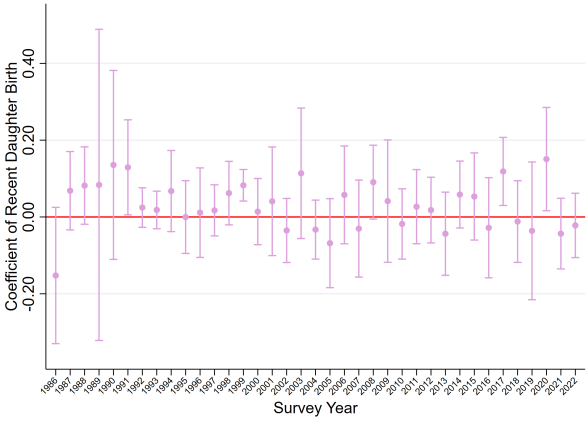
(a) Ideal Daughters



(b) Ideal Sons



(c) Ideal Total Children



Notes: OLS coefficient of recent daughter birth on stated ideal daughters, sons, and total children (Panels (a)–(c)), estimated separately for each survey year, analogous to Figure A.3 but splitting the sample by the year the interview took place rather than by respondent age. Each point = one survey year; capped lines = 95% CIs. Years with fewer than 1,000 observations are excluded. Specification: sibship dummies, age, age at first birth, education, rural, poor, marital status (unmarried/widowed/divorced, married omitted), age at first marriage, and child mortality count; region×interview-month fixed effects; population-weighted; robust SEs. Survey-year fixed effects are omitted here (unlike in the main specification) because the sample is already split by survey year, leaving no within-year variation for such a fixed effect to absorb. Variation across years indicates whether the estimated effect is stable over time or concentrated in particular survey waves.

A.8 Cultural and Macroeconomic Moderators

Table A.12: Macro Heterogeneity — Ideal Daughters

	(1) High lnGDPpc	(2) High Gender Eq.	(3) High Gini	(4) High HCI	(5) Dowry to Groom	(6) High Country TV
Recent Daughter Birth	0.0919***	0.1066***	0.0943***	0.1043***	0.0988***	0.0783***
	(0.0103)	(0.0105)	(0.0144)	(0.0079)	(0.0116)	(0.0093)
Macro/Cultural Indicator	-0.3387*	-0.2321**	-0.0633	-0.1679*	-0.6510***	-0.6891***
	(0.1755)	(0.0996)	(0.1196)	(0.0913)	(0.1650)	(0.1093)
Interaction Term	0.0316*	-0.0030	0.0136	0.0183	0.0018	0.0388**
	(0.0162)	(0.0119)	(0.0170)	(0.0228)	(0.0224)	(0.0172)
Constant	1.8955***	1.8565***	1.9238***	1.8539***	2.0813***	2.1579***
	(0.2548)	(0.2305)	(0.2380)	(0.2324)	(0.1761)	(0.1558)
Observations	207,869	207,869	207,869	207,869	153,232	202,367
R-squared	0.0924	0.0850	0.0774	0.0782	0.1337	0.1458

Notes: Each column interacts the treatment with a binary country-level macro/cultural indicator (above/below sample median). Columns: (1) ln GDP per capita, (2) Gender equality index, (3) Gini coefficient, (4) Human Capital Index, (5) Bride price flows to groom, (6) Country TV penetration. Because these moderators vary only at the country level, standard errors are clustered by country (82 clusters) rather than heteroskedasticity-robust, unlike the individual-level specifications elsewhere in the paper. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.13: Macro Heterogeneity — Ideal Sons

	(1) High lnGDPpc	(2) High Gender Eq.	(3) High Gini	(4) High HCI	(5) Dowry to Groom	(6) High Country TV
Recent Daughter Birth	-0.0663***	-0.0706***	-0.0608***	-0.0766***	-0.1046***	-0.0698***
	(0.0121)	(0.0132)	(0.0192)	(0.0104)	(0.0109)	(0.0095)
Macro/Cultural Indicator	-0.3515*	-0.2513**	-0.1126	-0.1880*	-0.6873***	-0.6756***
	(0.1834)	(0.1118)	(0.1434)	(0.1008)	(0.1527)	(0.1287)
Interaction Term	-0.0335	-0.0219	-0.0221	-0.0423**	0.0557***	-0.0340**
	(0.0240)	(0.0195)	(0.0200)	(0.0175)	(0.0191)	(0.0161)
Constant	2.1791***	2.1451***	2.2640***	2.1414***	2.4382***	2.4065***
	(0.2645)	(0.2372)	(0.2577)	(0.2393)	(0.2089)	(0.1753)
Observations	207,868	207,868	207,868	207,868	153,231	202,366
R-squared	0.1149	0.1065	0.0992	0.0996	0.1512	0.1562

Notes: See notes to Table A.12.

Table A.14: Macro Heterogeneity — Ideal Total Children

	(1) High lnGDPpc	(2) High Gender Eq.	(3) High Gini	(4) High HCI	(5) Dowry to Groom	(6) High Country TV
Recent Daughter Birth	0.0264**	0.0343***	0.0130	0.0320***	0.0075	0.0033
	(0.0132)	(0.0121)	(0.0176)	(0.0096)	(0.0118)	(0.0133)
Macro/Cultural Indicator	-0.4479	-0.4980**	-0.0926	-0.5462**	-1.3681***	-1.5330***
	(0.3855)	(0.2494)	(0.2390)	(0.2073)	(0.3297)	(0.2573)
Interaction Term	0.0087	-0.0127	0.0211	-0.0284	0.0282*	0.0260
	(0.0186)	(0.0172)	(0.0215)	(0.0210)	(0.0153)	(0.0174)
Constant	4.5075***	4.4130***	4.5299***	4.4091***	4.8677***	5.2815***
	(0.5741)	(0.5297)	(0.6019)	(0.5351)	(0.4805)	(0.4104)
Observations	247,819	247,819	247,819	247,819	180,523	227,445
R-squared	0.1532	0.1540	0.1439	0.1474	0.2137	0.2663

Notes: See notes to Table A.12.

A.9 Multiple Testing Correction

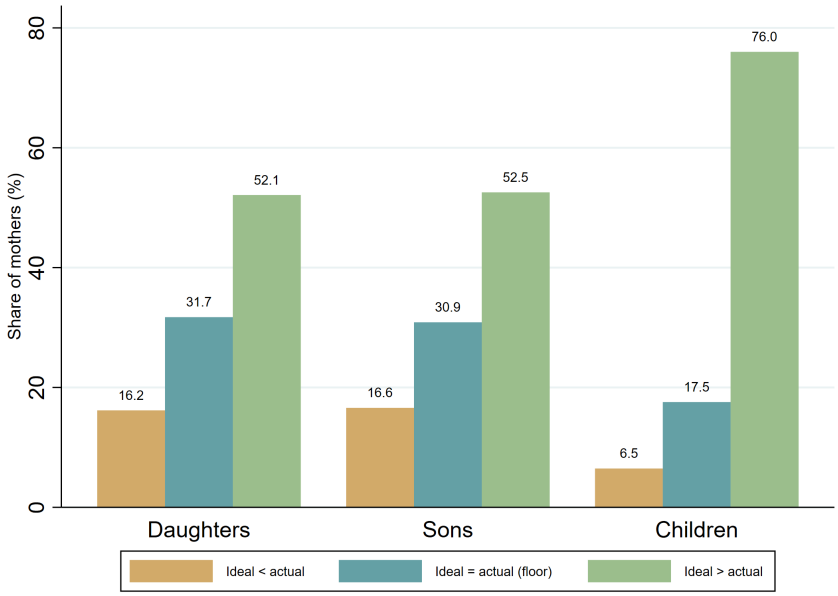
Table A.15: Benjamini-Hochberg Correction Across All Heterogeneity Tests

Domain	Outcome	Moderator	p	BH q
Individual	Ideal Sons	Rural	0.001	0.052†
Macro/Cultural	Ideal Sons	Dowry to Groom	0.004	0.063†
Macro/Cultural	Ideal Sons	Human Capital Index	0.016	0.188
Macro/Cultural	Ideal Daughters	TV Penetration	0.024	0.219
Macro/Cultural	Ideal Sons	TV Penetration	0.035	0.255
Macro/Cultural	Ideal Daughters	ln(GDPpc)	0.051	0.305
Macro/Cultural	Ideal Total Children	Dowry to Groom	0.064	0.331
Individual	Ideal Sons	Above-Avg. Education	0.123	0.491
Macro/Cultural	Ideal Total Children	TV Penetration	0.134	0.491
Individual	Ideal Sons	Experienced Child Death	0.137	0.491
Macro/Cultural	Ideal Sons	ln(GDPpc)	0.163	0.525
Macro/Cultural	Ideal Total Children	Human Capital Index	0.175	0.525
Individual	Ideal Daughters	Above-Avg. Age	0.201	0.539
Individual	Ideal Total Children	Poor	0.227	0.539
Macro/Cultural	Ideal Sons	Gender Equality	0.261	0.539
Individual	Ideal Total Children	Rural	0.263	0.539
Individual	Ideal Sons	Above-Avg. Age	0.269	0.539
Macro/Cultural	Ideal Sons	Gini Index	0.269	0.539
Individual	Ideal Daughters	Poor	0.319	0.588
Macro/Cultural	Ideal Total Children	Gini Index	0.327	0.588
Individual	Ideal Sons	Poor	0.370	0.633
Individual	Ideal Daughters	Above-Avg. Education	0.387	0.633
Macro/Cultural	Ideal Daughters	Gini Index	0.423	0.635
Macro/Cultural	Ideal Daughters	Human Capital Index	0.423	0.635
Macro/Cultural	Ideal Total Children	Gender Equality	0.460	0.640
Individual	Ideal Total Children	Above-Avg. Education	0.462	0.640
Individual	Ideal Total Children	Experienced Child Death	0.610	0.793
Individual	Ideal Daughters	Experienced Child Death	0.619	0.793
Macro/Cultural	Ideal Total Children	ln(GDPpc)	0.639	0.793
Individual	Ideal Sons	Above-Avg. Age at First Birth	0.710	0.852
Individual	Ideal Daughters	Above-Avg. Age at First Birth	0.750	0.871
Macro/Cultural	Ideal Daughters	Gender Equality	0.800	0.888
Individual	Ideal Total Children	Above-Avg. Age at First Birth	0.822	0.888
Individual	Ideal Total Children	Above-Avg. Age	0.838	0.888
Macro/Cultural	Ideal Daughters	Dowry to Groom	0.936	0.944
Individual	Ideal Daughters	Rural	0.944	0.944

Notes: All 36 interaction-term tests from Section 4.3.2 (Individual, Tables A.9–A.11) and Section 4.3.3 (Macro/Cultural, Tables A.12–A.14), pooled into a single family and ranked by raw p-value. BH q is the Benjamini-Hochberg step-down adjusted p-value; † marks tests surviving at a 10% false discovery rate. 0 of 36 tests survive at a 5% false discovery rate; 2 survive at 10%.

A.10 Floor Diagnostics

Figure A.5: Distribution of Ideal Relative to Actual



Notes: Grouped bar chart decomposing the distribution of the gap between stated ideal and actual family size, separately for daughters, sons, and total children. Bars represent: (i) share with stated ideal *below* actual count (tan: ideal $<$ actual), (ii) share at the floor where stated ideal equals actual (teal: ideal $=$ actual), and (iii) share with stated ideal *above* actual (light green: ideal $>$ actual). The floor share (ideal=actual) quantifies how binding the floor censoring constraint is. A behavioral preference shift should raise the gap *above* actual, not simply pile at the floor.

Table A.16: Treatment Effect Within Floor-Clean Strata

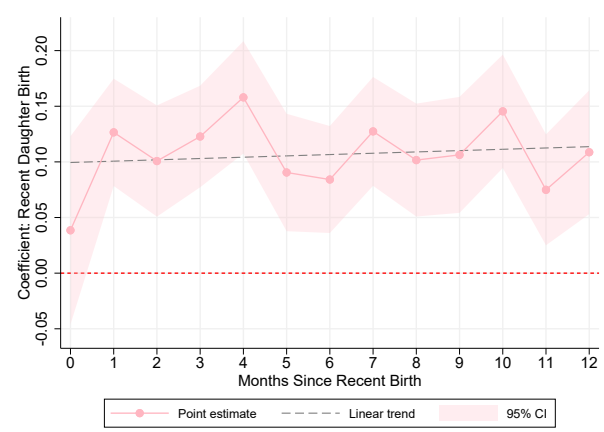
	(1) idG $\geq$2	(2) idG $\leq$-1	(3) idB $\geq$2	(4) idB $\leq$-1	(5) idK $\geq$2	(6) idK $\leq$-1
Recent Daughter Birth	0.8297***	0.5363***	-0.8062***	-0.5812***	0.0146	-0.0108
	(0.0195)	(0.0128)	(0.0186)	(0.0129)	(0.0157)	(0.0126)
Observations	52,743	38,063	56,747	39,173	138,935	17,238
R-squared	0.5454	0.5902	0.5526	0.6008	0.5009	0.7327

Notes: Each column restricts to a floor-clean stratum defined on the post-birth gap (ideal$-$actual) for the relevant outcome. *Gap* ≥ 2: women who stated an ideal at least 2 units above their current actual count. Because a single same-sex birth reduces the gap by exactly 1, these women remain above the floor even after the birth — floor censoring cannot mechanically affect their stated ideal. *Gap* ≤ -1: women whose stated ideal is already below their actual count; no upward floor pressure applies. Excluded are women with gap $= 0$ (at floor) and gap $= 1$ (marginal: a same-sex birth moves these women from above-floor into the at-floor stratum, creating endogenous selection between treatment arms). A significant treatment effect in either clean stratum constitutes evidence of genuine preference updating that cannot be attributed to floor censoring. Specification: full sibship controls; region$\times$interview-month and interview-year FE; population-weighted; robust SEs. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

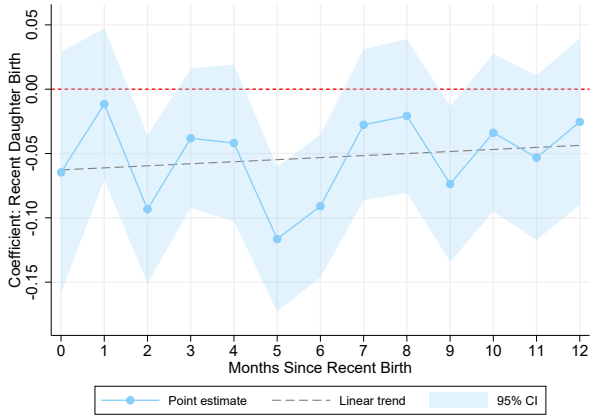
A.11 Timing Tests

Figure A.6: Month-by-Month Exposure Gradient

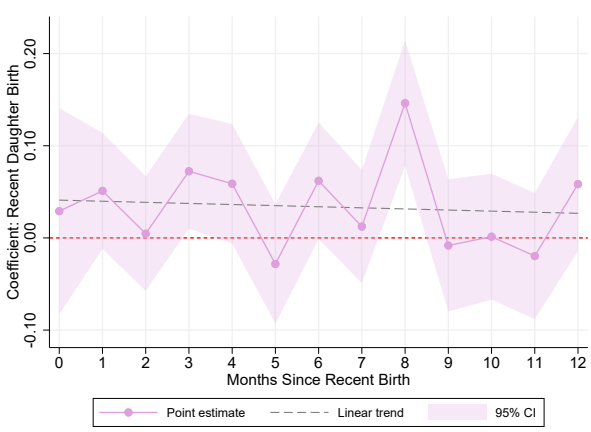
(a) Ideal Daughters



(b) Ideal Sons



(c) Ideal Total Children



Notes: Each point = OLS coefficient of a recent daughter birth on stated ideal daughters, sons, and total children (Panels (a)–(c)), estimated separately for each month since birth (among women with at least 200 observations). Dashed line = OLS linear trend across months; shaded band = 95% CI.

A.12 Household Contact

Table A.17: Summary Statistics: Household Contact Sample

	mean	sd	min	max	count
Focal mother					
Ideal Daughters	1.574	1.161	0	6	18,951
Ideal Sons	1.637	1.245	0	6	18,951
Ideal Total Children	3.756	2.046	0	12	18,951
Age	27.616	6.776	15	50	18,951
Education (Years)	6.183	4.642	0	21	18,951
Total Children	1.762	0.782	1	3	18,951
Rural Household	0.557	0.497	0	1	18,951
Poor Household	0.453	0.498	0	1	18,951
Married	0.542	0.498	0	1	18,951
Household					
Co-resident Recent Daughter Birth	0.489	0.500	0	1	18,951
Women Interviewed in Household	2.896	1.187	2	18	18,951
Other Women's Living Sons	1.177	1.671	0	23	18,951
Other Women's Living Daughters	1.235	1.766	0	15	18,951
Co-resident's Parity	1.681	0.769	1	4	18,951
Months Since Co-resident's Birth	5.971	3.632	0	12	18,951
Pair: Co-wife	0.090	0.287	0	1	18,951
Pair: In-law	0.334	0.472	0	1	18,951
Pair: Blood Relative (collateral)	0.260	0.439	0	1	18,951
Pair: Other / Unrelated	0.315	0.464	0	1	18,951
Observations	18951				

Notes: Summary statistics for the full-specification estimation sample in Table 5. Unweighted. Focal mother rows describe the non-birthing mother; household rows describe her household and the co-resident's recent birth. Pair-type indicators classify the relationship between the focal mother and the co-resident birth mother.

A.13 PSM Specification

I use propensity score matching (PSM) as a further robustness check to my OLS results. In order to ensure that the key assumptions of matching (ignorability and common support) are satisfied, I first check initial balance amongst the covariates. Table 2 provides an analysis of balance of covariates between treatment and control. Relevant, observed covariates in the dataset are age at first birth, years of education, rural or urban status, poverty status, interview month, interview year, marital status, age at first marriage, child mortality count, and the seven pre-existing sibship dummy variables. Of these covariates, the dummy indicating the recent birth is the mother's first child, two of the pre-existing sibship dummies (Son-Daughter and Daughter-Daughter), widowed status, and total child mortality are imbalanced at the 5% level; age at first birth, poor households, and interview year are not significant. These imbalances concentrate in pre-existing sibship composition and child mortality, which is exactly why the primary specification conditions on sibship cell and separately investigates child mortality as a possible signal of sex-selective childcare (Section 4.3.2); sensitivity analyses on the matching specifications are nonetheless warranted. While the balance of covariates is imperfect, there is an extensive region of common support, as displayed in Figure A.1. The propensity score distributions are nearly identical, which is encouraging.

The specific PSM estimator of choice is Nearest Neighbor PSM with no caliper, matching on the mother's age, age at first birth, years of education, rural and poor households, interview month, interview year, and the seven pre-existing sibship dummies. I want to capture as much variance in the covariates as possible and introduce as little bias as possible, so only the nearest neighbor is used (Austin, 2010). All treated observations are able to be matched to untreated observations.

Table A.18: PSM Average Treatment Effect on the Treated

	(1) Ideal Daughters	(2) Ideal Sons	(3) Ideal Total Children
ATET			
ATT of Recent Daughter Birth	0.0877***	-0.0939***	-0.0091
	(0.0060)	(0.0064)	(0.0095)
Observations	243,182	243,181	286,580

Notes: Propensity score matching using nearest-neighbor matching (one match, with replacement) on the logit propensity score. Propensity score covariates: age, age at first birth, education years, rural, poor, interview month, interview year, and the seven pre-existing sibship dummies. The ATT (average treatment effect on the treated) reports the mean difference in stated ideal family composition between matched treated and control women. Abadie–Imbens standard errors; unweighted. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.14 IPWRA Specification

IPWRA continues to be a consistent estimator even in the presence of misspecification of either the treatment or the outcome model, though not both (Wooldridge, 2010).

My use of IPWRA maintains identical covariates to those of my PSM specification: the mother's age, age at first birth, interview year, interview month, years of education, dummies for rural households and poor households, and the seven pre-existing sibship dummies. Inverse probability scores of assignment to treatment are calculated using a logit model with these covariates, with treatment being if a given mother had a daughter birth in the past 12 months, and zero if they had a son instead. Likewise, the outcome variables are the mothers' stated ideal number of daughters, sons or total children.

Table A.19: IPWRA Average Treatment Effect on the Treated

	(1) Ideal Daughters	(2) Ideal Sons	(3) Ideal Total Children
ATET			
ATT of Recent Daughter Birth	0.1028***	-0.0837***	0.0222***
	(0.0045)	(0.0048)	(0.0071)
Observations	243,182	243,181	286,580

Notes: Inverse-probability-weighted regression adjustment (IPWRA). Propensity (treatment selection) model: logit on age, age at first birth, education years, rural, poor, interview month, interview year, and the seven pre-existing sibship dummies. Outcome model: linear regression. The doubly-robust IPWRA estimator is consistent if *either* the propensity or outcome model is correctly specified. ATT = average treatment effect on the treated. Robust SEs; unweighted. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.